

Improving Learning

RANZCP Examination Review

Final Report

Submitted to the Royal Australian and New Zealand
College of Psychiatrists (RANZCP)

Australian Council for Educational Research

January 2020

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ACKNOWLEDGEMENTS

We would like to formally acknowledge the support of the Royal Australian and New Zealand College of Psychiatrists throughout this review. In particular, we express our gratitude to College staff, the Examinations Committees, and Examiners. Examiners and subcommittee members were welcoming and responded openly and thoughtfully throughout the review. All other stakeholders gave their time freely, responded with considered insights and were very accommodating and helpful.

EXECUTIVE SUMMARY

BACKGROUND

The Australian Council for Educational Research (ACER) was formally engaged in July 2019 to conduct a full-scale assessment review for the Royal Australian and New Zealand College of Psychiatrists (RANZCP). The aim of this work was to provide recommendations for improvements to examinations and examination processes in the College to ensure that examinations and assessments are reliable, valid, of high quality and as efficient as practical.

The review was commissioned in response to a RANZCP Board initiative to address consistently low pass rates in the Essay-style examination, and in response to ongoing concerns from trainees and other stakeholders about the perceived challenge of passing the Essay-style examinations. Although these concerns were the starting point for the review, the review scope included three components of the Fellowship Examination currently administered by the RANZCP: the Multiple-Choice Question (MCQ) examination, the Essay-style examination, and the Objective Structured Clinical Examination (OSCE).

The review process involved, where relevant, an analysis of the development, delivery, marking, reviewing and reporting of assessments, as well as standard setting processes and cut-score determinations, support for candidates, and training and calibration for examiners. The project was thus conceptualised as a 'gap analysis', whereby existing assessment procedures were compared with best practice in assessment in order to identify recommendations for improvement and enhancement of existing processes. The overarching focus for recommendations was to suggest areas for improvement that would ensure examination processes were transparent, consistent, fair and ultimately, defensible.

The project took place between July and December 2019 and comprised six phases: (i) project initiation, (ii) resource gathering, (iii) desktop review, (iv) observation and consultation, (v) data analysis from RANZCP surveys, and (vi) development of the final report with recommendations. The findings of our assessment reviews are underpinned by several key concepts drawn from the contemporary literature on assessment, and particularly in relation to medical education and speciality training. These are: (i) assessment and learning, (ii) aligned and fit-for-purpose assessment, (iii) collaborative construction of standards, (iv) defensibility of professional judgements, (v) fairness and consistency, and (vi) limits and feasibility of assessments.

SUMMARY OF KEY FINDINGS

Overall findings

All aspects of the RANZCP Examinations were interrogated in a thorough and robust manner. High-stakes examinations are sensitive to external review, and we were grateful for the high level of engagement and openness we experienced from RANZCP stakeholders.

- For the great majority of individuals involved at different stages of the examination process, there was a strong commitment to educational improvement and best-practice in assessment.
- The relevant committees and the College should be commended on their commitment to developing and delivering high-quality examinations; however, there are many aspects of the examinations and examination processes that require modifications to align with best practice.
- Examination processes have evolved organically over many years, generating somewhat convoluted implementations of certain methodologies. Examination processes for RANZCP need to be streamlined and simplified.

- At present, there are many areas that would be subject to scrutiny and pose a risk to the integrity of examination processes at the College. These have been highlighted throughout the report and form the basis of our recommendations.
- While there was general agreement that the majority of candidates who pass the Fellowship Examinations are safe to practice at a basic level, a substantial issue was identified: many stakeholders believed that the three examinations together do not assess all core psychiatric skills.
- The focus of the examinations may not be completely aligned with the learning outcomes of the training program. Core skills are possibly assessed outside of the current examinations, but assessing them in a lower-stakes context impacts trainee learning.

MCQ Examination findings

Key findings related to the MCQ Examination in terms of its purpose and format, item quality, constructive alignment, and fitness-for-purpose are as follows:

- The purpose was well understood by all relevant stakeholders as a test of basic knowledge.
- There was some concern that although the test is supposed to be aimed at a junior consultant level, trainees have not actually attained this level of competency when they undertake the exam. Most stakeholders agreed that the examination was not well targeted to an appropriate standard, partly because trainees can sit the examination after only 6 months of FTE training.
- Although several technical flaws were detected in the EMQs, most items provided an authentic clinical task, a focused lead-in, and a coherent set of plausible options. However, the current practice of using broken-up EMQs is likely to affect the authenticity, level of difficulty, and ultimately the quality of the MCQ exam.
- There were some concerns that the items were too easy, 'dumbed down', and could be passed using test-wise strategies. The EMQ format was seen to be suboptimal, and committee members were interested in exploring options for item reform.
- The Critical Analysis Problems (CAPs) appear to present authentic and meaningful tasks for candidates. However, some questions are factual in nature and are not reflective of the purported critical nature of the task. Further, some CAPs use a cumbersome 'fill-in-the-blank' format that is less authentic than the standard Single Best Answer format.
- Blueprinting processes to the item level exist for the MCQ Examinations; however, no assessment framework was sighted. This raises concerns about the robustness of the alignment between the examination and the learning outcomes of the training program.
- The general consensus was that the MCQ examination was fit-for-purpose as an exam that assesses fundamental background knowledge. However, from an educational perspective, there is certainly room for improvement. The focus could be honed to the fundamentals of psychiatric training and the examination could become a hurdle to training progression.

Essay-style examination findings

Key findings relating to the Essay-style examination in terms of its purpose and format, item quality and rubric design, constructive alignment, and fitness-for-purpose are as follows:

- The examination is clinically focused and is designed to assess critical thinking in relation to clinical practice and broader issues relevant to the psychiatric profession, including sociocultural and ethical issues, as well as the application of clinical knowledge.
- The Modified Essay Question (MEQ) marking guides are well-structured, separating and outlining different components to be judged and scored for each sub-question. However, some appear somewhat underdeveloped. Explicit criteria for each level, including which score or code to assign to a response, is missing for most of the questions.

- The analytical structure and description of the various levels in the Critical Essay Question (CEQ) marking guides within each competency is commendable. However, some were too generic and missing example responses.
- There were suggestions that, in practice, examiners find the marking guides difficult to use. There appears to be considerable room for variation in judgement between markers. These could be improved by developing robust principles-based rubrics that are task-specific.
- The smaller number of items and more in-depth focus on testing higher-order thinking in the Essay-style examination places limits on the breadth of sampling. Although items are mapped to competencies, there was no framework that provided guidelines to examiners around appropriate curriculum coverage in any one administration.
- Although the Essay-style examination is important in assessing whether trainees can demonstrate their awareness of social and cultural issues, the lack of an assessment framework that clearly identifies the link between what is being assessed and the learning outcomes of the broader training program is concerning. Specifically, it is unclear how the skills being assessed under the examination conditions in the CEQ reflects the real-world practice of being a psychiatrist.
- The rationale for the weighting of marks (40 for CEQs and 180 for MEQs) was queried by many of those interviewed. It was unclear why the CEQs and MEQs were aggregated at all as they appear to be functioning as two separate assessments.
- The low pass rates on the Essay-style examination were a concern to many. This examination appears to be more difficult for trainees with non-English speaking backgrounds.
- Trainees, SMIGs, Directors of Training and Supervisors were generally less favourable in their responses to issues concerning this examination than for the MCQ Examination.

Written Examination processes findings

Key findings relating to the Written Examination item development, exam delivery, marking processes, psychometric analyses, standard setting, borderline candidate review, training and calibration of examiners, support for candidates, and candidate reporting, are as follows:

- Although they have evolved organically over time, the processes for drafting, selecting and reviewing items appear robust. The blueprint appears to be the starting point for item development, which occurs in a collaborative and iterative fashion.
- Limitations with the exam delivery platform have led to inauthentic presentations of content in the MCQ examination due to inability to cope with the EMQ item type.
- The Essay-style examination is delivered by paper, which makes marking difficult, as markers must decipher candidate handwriting. It would be worth further exploring the options for moving to an online platform to deliver this examination.
- Although the Essay-style examination is double marked, the threshold for marking discordances is currently set too high. This has the potential to impact results from borderline candidates. Electronic marking would also remove the potential of many human errors and enhance the efficiency of the marking process.
- The comprehensive MCQ psychometric item analysis was encouraging. However, it would be possible to standardise the item analysis output reports and to enhance its aesthetic appeal. Further detailed analyses, for example of Item Characteristic Curves (ICCs) and Differential Item Functioning (DIF), would also help to optimise the fairness and consistency of the written examinations.
- The current manual approaches implemented by the college to the formatting, storage and processes of assessment data are risky and require an overhaul.
- Although MCQ item statistics are provided to the committee for review, many panel members noted that they had a limited understanding of the analyses. As many are not confident on

how to interpret these data, some targeted training for committee members would be worthwhile.

- The processes for weighting within criteria for score calculations are somewhat arbitrary and add an unnecessary layer of complexity to the scoring process. There are also flow on effects for the validity of the standard setting. Weighting of the relative coverage of certain knowledge and skills ought to be addressed through the blueprinting entirely.
- There was strong agreement that the Ebel method used to standard set the MCQs is highly convoluted. The current implementation has evolved organically over time and has become overly complex.
- Although the standard setting methodologies employed for the Written examinations are thorough and detailed, the current approaches should be either simplified or completely replaced by a simpler method.
- The processes for reviewing candidate results appears robust and thorough. However, there are no explicit criteria for determining which candidates fall into the borderline range for further review nor for determining which candidates might potentially be awarded a concessional pass.
- Although there is no formal training pathway for new examiners, informal training and induction does occur. It was generally acknowledged that more could be done in terms of training and support for both new and experienced examiners.
- Candidates are given access to a wide range of resources to help them prepare for their examinations. However, more collaboration between the training networks around making material available to other trainees and SMIGs, especially those placed in regional and remote centres, would be worthwhile.
- In their results letters, candidates receive their percentage scores by examination component as well as by blueprint domain. Although this is good practice, there were some concerns about the validity of the domain-level reporting due to the quality of the blueprinting.

OSCE findings

Key findings relating to the OSCE purpose and format, station quality and rubric design, constructive alignment and fitness-for-purpose are as follows:

- The OSCE aims to provide a robust way of ensuring that trainees are ready for independent practice by providing an opportunity to assess across a range of domains and interactions, and garner multiple examiner perspectives on a candidate.
- There were concerns raised that the large number of stations and short time available resulted in assessment of a very narrow spectrum of practice; indeed, as is the case with most OSCEs, each station only allows for a cursory assessment of psychiatric practice rather than a deeper probing of the art of psychiatry.
- Overall the OSCEs appear to be appropriately used and written, providing authentic clinical contexts and tasks for candidates, with relatively clear marking guides tailored to both clinical context and domain. However, there was a significant inconsistency within the Category 3 'Achieves Standard' descriptors that should be clarified, and a couple of relatively minor discrepancies in other score categories.
- Although the OSCEs are blueprinted by station to Descriptor Categories, Areas of Practice, CanMEDs Domains and Fellowship Learning Outcomes, there was no overarching assessment framework that provided guidelines to the subcommittee about appropriate coverage of content and other domains in any one administration. There was also a lack of detailed blueprinting by station criteria.
- Although the OSCEs appear fit-for-purpose, there was widespread disquiet that the OSCEs were not assessing the core skills of psychiatrists. This speaks to a deeper philosophical

question about the fitness-of-purpose of the OSCEs and whether the purported purpose is valid. As they stand, the OSCEs are fit-for-purpose in that they are clearly assessing what OSCEs are supposed to assess. However, this does not mean that all core areas of psychiatric practice are currently assessed by the OSCE.

- One theme that arose overwhelmingly throughout the consultations was that core skills around meeting and communicating with real patients, taking a history, conducting a patient examination, making a diagnosis and formulating a treatment plan, are not being examined. There was strong consensus that assessing these skills only in a formative setting has negatively impacted the training program.

OSCE processes findings

Key findings relating to the OSCE station development, exam delivery, marking processes, psychometric analyses, standard setting, borderline candidate review, training and calibration of examiners, support for candidates, and candidate reporting, are as follows:

- The overall development process for the OSCEs is collaborative and iterative and goes through multiple versions prior to completion. This process appears robust.
- Delivery of the OSCEs is an extremely complex exercise involving significant numbers of examiners, candidates and College staff. This is very well managed and deployed. However, the increasing numbers of candidates presents a long-term challenge for College resources and volunteer examiners.
- The lack of video-recording of the OSCEs poses a risk to the college in the case of legal appeals. As well as enhancing options for training and calibration, recording examinations would ensure that objective evidence exists in the case of a legal challenge.
- It would be worth exploring the options for moving to an electronic (such as a tablet) marking platform. This would streamline the marking process and reduce the potential for errors.
- The rationale and processes for converting domain ratings (from a categorical scale) to numerical scores is questionable. From a measurement perspective, multiplying the scores by proportionality factors to arrive at a final score out of 50 seems arbitrary, and runs the risk of losing the essential assessment information contained in the examiner ratings associated with assessment domains.
- Significant data is being collected on the different marking domains, but this data is currently not being aggregated and reviewed to interrogate patterns of performance across domains.
- The current approach to standard setting the OSCEs (Contrasting Groups or Borderline Groups) is a defensible performance-based method, although its precise implementation requires careful reconsideration. The raw categorical data from the marking rubric is subjected to a convoluted transformation process to arrive at the final score, which undermines the validity of the standard setting process. Examiners were equally concerned about the way the standard setting is implemented; most of the disquiet arose due to perceptions that candidates were passing overall who should not be passing.
- The current model for determining the overall cut score is a compensatory model. This focuses on overall performance, aggregating scores across all stations in the exam. The holistic rubric approach with its task specificity suggests a model whereby minimum standards need to be met across the curriculum and blueprinted domains (i.e., a conjunctive model). Currently, candidates may be meeting the overall cut score but performing consistently below the acceptable standard in certain blueprint categories.
- While the process for borderline candidate review appears robust and thorough, no specific criteria exist to consistently identify candidates in the borderline range. Far more data could be made available to the panel to aid deliberations. For instance, where small deficiencies in

performance in certain domains are identified, other evidence from across the assessment program could be used to inform decision-making.

- While there is no formal training pathway for new examiners, informal training and induction does occur. An impressive amount of work does go into the training and calibrating of examiners prior to the OSCE. The calibration work done the day before the OSCEs aligns with best practice.
- Candidates are given access to a wide range of resources to help them prepare for the OSCEs. However, more collaboration between the training networks around making material available to other trainees and SMIGs, especially those placed in regional and remote centres, would be worthwhile.
- In their results letters, candidates receive their specific scores by station as well as by blueprint domain. Although this is good practice there were some concerns about the validity of this as the data points in each category are not disclosed.

Other findings relating to the assessment program

Finally, several key findings arose relating to the broader program of assessment. These are as follows:

- As is the case in many colleges, trainees who fail examinations have often done poorly in their formative assessments, but no intervention was undertaken to rectify their gaps in knowledge and ability. This points to a clear disconnect between the relevant examination and training committees.
- The clear-cut distinction between purely formative and summative assessment is unhelpful and perpetuates a false-dichotomy. Programmatic thinking that bridges the gap between training and assessment should be incorporated into the training and assessment programs.
- The examinations appear to be driving trainee learning, as they automatically become eligible for these after fixed times. For example, many stakeholders believed that some trainees were sitting their OSCEs too early in training, before they had reached the required level of competency.
- Although training is articulated as a 'competency-based' fellowship program, the training pathway appears to be functionally time-based in that progression occurs automatically based on number of months in FTE training. This does not align with the fact that trainees learn and progress at different rates in different domains.

COMMENDATIONS

Before presenting our recommendations, we would like to note many commendable aspects of the existing RANZCP examination processes:

- The examiners and relevant committees demonstrated a strong commitment and dedication to developing high-quality examinations. They were also open to educational input and willing to modify directions and expectations based on open dialogue.
- The blueprinting processes implemented (particularly for the written examinations) to ensure constructive alignment are thorough, considered and robust. The detail recorded at the item-level and the reporting potential this provides is notable.
- Processes for the collaborative, iterative development of new items for the written examinations appear robust. While some avenues for enhancement have been noted below, the detail and descriptions in the marking guides for the Essay-style examinations ensure that the rubrics are sound.
- The augmentation or replacement of the previous classical test theory statistics (and some simplistic thresholds) with Rasch test and item statistics will provide more comprehensive

information for scrutinising the fitness-for-purpose and quality of the exam and individual items.

- The OSCE stations appear well-thought out, provide authentic clinical contexts and tasks for candidates, with relatively clear marking guides tailored to both clinical context and domain. The skills assessed are appropriate and consistent with the kinds of skills typically assessed by OSCEs in medical and health education contexts.
- The clear attempt in the OSCEs to tailor the descriptions of minimum standard to the specific scenario is highly commendable and would appear to significantly support the validity of such decisions. This principled approach recognises that candidates may achieve the standard in slightly different ways, while still meeting this core criterion.
- Delivery of the OSCEs is an extremely complex exercise involving significant numbers of examiners, candidates and College staff. The smooth operation of these examinations was impressive.
- The efforts made to calibrate examiners for the OSCEs, tied to the training sessions the day before the exam, were high-quality.
- Although areas for improvement have been identified, the process for reviewing candidate and cohort results for all examinations was thorough.
- The amount of material made available to trainees to support them in their examination preparation is commendable.
- It was pleasing to see reports for Directors of Training (with specific mention if a trainee requires targeted learning or to show cause), Branch Training Committees, reports at the State level, as well as a detailed post examination report from the Committee for Examinations.

RECOMMENDATIONS

As a result of the review, the following recommendations should be considered to improve the efficiency, effectiveness and rigour of the RANZCP Examinations. Some recommendations relate to global or 'big-picture' reform, while some relate to process specific reform. For each recommendation, there are some proposed specific actions for the college to consider. Recommendations are also presented in terms of their urgency. In the context of high-stakes examinations increasingly being subjected to scrutiny, we encourage the College to carefully reflect on these suggestions and take them into account.

Urgent recommendations	Proposed actions
<i>U1. Ensure better integration between training and assessment.</i>	• Develop assessment frameworks for the examinations that explicitly demonstrate the alignment between assessment moments and the learning outcomes of the training program. • Conduct a thorough review of the efficacy of the workplace-based assessments (WBAs), Entrustable Professional Activities (EPAs) and In-Training Assessments (ITAs). In doing so, look at incorporating programmatic thinking into decision-making around trainee progression, such as triangulating data from lower-stakes assessments in a meaningful way to inform review panels. • Explore options for either reintroducing a form of the Observed Clinical Interview (OCI) examination (which may subsequently reduce the number of stations in the OSCE), or triangulating assessment data from the Observed Clinical

	Assessments (OCAs) in a way that encourages a positive educational impact for trainees. Encourage more collaborative dialogue between the Committee for Examinations and the Committee for Training.
<i>U2. Align the MCQ examination with the early parts of fundamental training.</i>	- Review the purpose and expected standard of the MCQ examination to better align with the early parts of training. - Adapt the existing MCQ items to ensure they are fit-for-purpose. For example, re-aggregate the EMQs as originally written, or edit each item to have a more coherent and plausible set of 4 or 5 options.
<i>U3. Reform the content and format of the Essay-style examination to better align with its purpose.</i>	- As they appear to be assessing different skills, separate the CEQ and MEQ components of the Essay-style examination and score them as independent assessment elements. As part of this, ensure that the format adequately aligns with the purpose of the examination, and modify if necessary. - Enhance the CEQ and MEQ marking guides to include more task-specific information relating to poor or unacceptable answers and the underlying rule or principle to enable grading of unanticipated or unspecified candidate responses. Including such information for markers (where appropriate) would likely optimise the reliability, discrimination and effectiveness of the marking process.
<i>U4. Establish consistent criteria for post-examination review and ensure progression decisions are based on a robust evidentiary framework.</i>	- Consider using standard error calculations from each examination component separately (e.g., EMQ, CAP, MEQ and CEQ) to establish criteria that can be applied when reviewing results for candidates falling into the borderline range and determining which candidates might potentially be awarded a concessional pass. These criteria should be applied in conjunction with a broader evidentiary framework that also considers evidence from other assessments. - When reviewing borderline candidates in the post-examination review, deliberations by the panel should focus on available data using an evidentiary framework. Other evidence from across the assessment program could be used to inform decision-making.
<i>U5. Develop progression rules for examination eligibility through phases of training.</i>	- Explore options for a hybrid model of training that implements competency-based hurdles, ensuring that trainees progress to the next phase of training when they have demonstrated adequate competency. - For example, outline that trainees need to pass the MCQ exam before proceeding to the Essay-style examination, then must pass both components of the Essay-style examination independently before progressing to the OSCEs.
<i>U6. Review the rationale and need for weighting</i>	Considerations around appropriate coverage of curriculum domains should be adequately addressed through well-

<i>components throughout all examinations.</i>	constructed blueprints and focussing on essential elements of the training program when developing items.
<i>U7. Reform the standard setting methodologies for all examinations.</i>	• Review the precise methodologies being deployed to standard set the written examinations. The methods should be simplified or replaced with a more straightforward item-based method. • Establish conjunctive rules for having to pass a certain number of OSCE stations. Trial this with real data to ensure that an empirical evidence-base informs the development, and the impact on the cohort should be analysed prior to finalising and publishing conjunctive rules. • Explore options for aggregating OSCE data across stations by other marking domains (such as CanMEDs and other assessment categories) to establish conjunctive rules by domain. This would incorporate programmatic thinking into the OSCEs and ensure minimum standards were being met across stations. However, the feasibility and reliability of aggregating OSCE data in this way would need to be investigated empirically prior to implementation.

Short-term recommendations	Proposed actions
<i>S1. Enhance the efficacy of the delivery and marking processes for the written examinations</i>	• Explore options for a more flexible computer-based examination platform that can manage a range of item types and deliver the Essay-style examination. • Explore the feasibility of moving to an online platform for marking. This would remove the potential of many human errors and would enhance the efficiency of marking processes. • Lower the discordance threshold for resolving the double marking in the Essay-style exam, especially for candidates whose final average mark is close to the cut score. • Explore the feasibility of giving examiners more time between candidates to make some specific notes that could be collated and provided as specific feedback to candidates. The provision of electronic (tablet) marking would streamline this process.
<i>S2. Simplify the OSCE marking design and scoring processes.</i>	• Review the rationale and process for converting domain ratings (from a categorical scale) to numerical scores. Ensure that this translation adds value rather than clouding the rich raw data. • Ensure that inconsistencies in the Category 3 'Achieves Standard' descriptors are adequately clarified, and review minor discrepancies in other score categories.

<i>S3. Enhance the psychometric approaches to the examinations.</i>	- Integrate the psychometric analysis with software for report production (e.g., R) to enable highly automated and professionally formatted technical reporting for all examinations. - Conduct further psychometric analyses to optimise fairness and consistency, such as ICCs and DIF analysis. - Produce detailed technical documentation on each test administration.
<i>S4. Formalise examiner training pathways and ensure that examiner training is highly relevant.</i>	- Establish formal training pathways for new examiners (for both Written and OSCE exams) and provide more ongoing training and support around best-practice in assessment for current examiners. - Deliver training on item-writing best practice and ensure that item writers maintain quality through expert panel forums. - Deliver targeted training for committee members on the theory and practice of psychometric analysis to ensure that relevant stakeholders fully appreciate the utility of the available psychometric data.
<i>S5. Enhance feedback to candidates in all candidate reports.</i>	Explore options for giving counts of items by blueprint domain and show candidate location relative to the cut score (even for successful candidates).
<i>S6. Ensure all trainees and SIMGs have access to up-to-date support material</i>	Encourage more collaboration between training networks to ensure examination support material is available to trainees, especially for those placed in regional and remote centres.

Longer-term recommendation	Proposed actions
<i>L1. Invest in IT infrastructure to streamline content management, buttress security, and ensure data integrity.</i>	- Explore the options for utilising new item banking software and investing in IT infrastructure to ensure item security, secure file transfer mechanisms, and monitor version control. Item banking platforms can be used to streamline the entire test development and delivery process for both the Written and OSCE examinations. - Improvements to the format, storage and processing of assessment data would be possible across the examination program. Specifically, calculations and annotations should be separated from raw data tables in spread sheets. In addition, protocols for processing scoring key adjustments or question deletions should be reviewed to ensure that they avoid requiring unnecessary manual changes to raw score data.
<i>L2. Consider filming all OSCEs.</i>	Explore options for filming OSCEs both in hospitals and built-for-purpose examination centres. This footage could be used for future training and calibration purposes, and provide important objective evidence in the case of candidate appeals.

INTRODUCTION & BACKGROUND

OVERVIEW

The Australian Council for Educational Research (ACER) was formally engaged in July 2019 to conduct an assessment review of selected Fellowship Examination assessments for the Royal Australian and New Zealand College of Psychiatrists (RANZCP).

Evaluative reviews such as this are conducted regularly in medical education contexts and reflect good practice in seeking continuous improvement of assessments. They can also be helpful in clearly identifying areas that need focused attention between external accreditation cycles. The aim of this work was to provide recommendations for improvements to examinations and examination processes in the College to ensure that examinations and assessments are reliable, valid, of high quality and as efficient as practical.

The review was commissioned in response to a RANZCP Board initiative to address consistently low pass rates in the Essay-style examination, and in response to ongoing concerns from trainees and other stakeholders about the perceived challenge of passing the Essay-style examinations. Although this was the starting point for the review, the scope included three components of the Fellowship Examination currently administered by the RANZCP: the Multiple-Choice Question (MCQ) examination, the Essay-style examination, and the Objective Structured Clinical Examination (OSCE).

OBJECTIVES AND FOCUS

The RANZCP is accredited by the Australian Medical Council (AMC) as the specialist medical college responsible for delivering education, training and professional development programs for psychiatrists in Australia and New Zealand. RANZCP conducts a rigorous psychiatry training program that takes a minimum of five years to complete. During this time, trainees work in hospitals and clinics under the supervision of experienced psychiatrists.

The training program, completed in a minimum of 60 months full-time equivalent (FTE), aims to produce high-quality medical specialists with a range of specialised knowledge and complex medical skills. These professionals also work in high-risk contexts where mistakes can have drastic consequences for patients. As such, specialised training programs are increasingly subject to scrutiny.

In order to become a Fellow of the RANZCP, trainees must meet the requirements of the course as set out in the Training Program documents. Training in the Fellowship Program is completed in three stages: Stage 1 (12 months FTE), Stage 2 (24 months FTE) and Stage 3 (24 months FTE). Progression through these stages is evaluated through a series of assessments that are part of the training process, to provide feedback to trainees on their development. In addition, trainees and partially comparable Specialist International Medical Graduates (SIMGs) complete College-administered assessments, which comprised the core focus of this assessment review. As part of the Fellowship Examination, candidates undertake three high-stakes examinations:

- A Multiple-Choice Question (MCQ) examination, comprised of 140 Extended Matching Questions (EMQs) and 2 Critical Analysis Problems (CAPs) (of 190 minutes duration)
- An Essay-style examination, comprised of one Critical Essay Question (CEQ) and 4–6 Modified Essay Questions (MEQs) (of 180 minutes duration)
- A 15-station (11 active, 4 byes) Objective Structured Clinical Examination (OSCE) (of 200 minutes duration).

For this examination review project, we conducted an initial review of these three components of the Fellowship Examination currently administered by the RANZCP. The examination review process involved, where relevant, an analysis of the development, delivery, marking, reviewing and reporting of assessments, as well as standard setting processes and cut-score determinations, support for candidates, and training and calibration for examiners.

The project was thus conceptualised as a 'gap analysis', whereby existing assessment procedures were compared with best practice in assessment in order to identify recommendations for improvement and enhancement of existing processes. The overarching focus for recommendations was to suggest areas for improvement that would ensure examination processes were transparent, consistent, fair and ultimately, defensible.

METHODOLOGY

The project took place between July and December 2019 and involved the six phases described below.

Phase 1: Project initiation

- Defined terms of reference and scope of the review.

Phase 2: Resource gathering

- Resources gathered from RANZCP for the review included documentation relating to the development and delivery of the different examinations, to examination and assessment procedures, previous exams, examination data, examiner reports, marking rubrics/scoring keys, blueprinting information, and information for candidates about examinations.

Phase 3: Desktop review

- ACER reviewed assessment materials submitted for the review.

Phase 4: Observation and consultation

- Attended the Examiner and Role Player training for the OSCEs in Perth on 13 September 2019.
- Observed the OSCEs in Perth on 14 September 2019.
- Undertook face-to-face discussions with examiners and College staff during meetings and observation sessions where feasible.
- Conducted 10 semi-structured phone interviews with relevant stakeholders, including Chairs and Deputy Chairs of examination committees.
- Considered written submissions from other subcommittee members.
- Met with College staff and viewed material available to trainees on the Learnit system on 21 October 2019.

Phase 5: Data analysis from RANZCP surveys

- RANZCP developed and administered surveys of trainees, Directors of training and supervisors, and the Committee for Exams in parallel with this review project. Summary data from these surveys were provided to ACER. In particular, quantitative data is referred to throughout this report.
- Trainees (89%) and Specialist International Medical Graduates (11%) completed a survey developed and administered by the College about their perceptions of the College examinations. Four hundred and sixteen respondents from all states and territories of Australia commenced the survey, but not all respondents answered all questions. Most respondents worked fulltime (84%).

- Directors of training and supervisors from all states and territories of Australia responded to a 20-item survey assessing their perceptions of the use of the Multiple-Choice question (MCQ) examination as part of the training program (between 190 and 205 responded to each item).
- Fifteen members of the Committee for Exams responded to questions regarding the quality of the three examinations.

Phase 6: Development of final report with recommendations

- ACER synthesised the information from document reviews, consultations, notes from attending meetings and observing examinations, and survey data supplied by RANZCP to develop recommendations for areas identified for development.

EVALUATIVE FRAMEWORK

The findings of the assessment reviews undertaken by ACER are underpinned by several key concepts drawn from the contemporary literature on assessment, and particularly in relation to medical education and speciality training. The definitions of each concept and the approaches taken in applying them are described in the sections below.

Assessment and learning

This principle ensures that assessment and learning are integrated. Each assessment moment should have a clear educational value. All assessment should provide rich, meaningful and timely feedback to students and provide feedback on performance that contributes to students' learning (Sadler, 1998). Data can be used by supervisors/trainers to identify gaps in learning and to help them provide targeted teaching (Bennett, 2011). All assessments (both low- and high-stakes ones) can provide useful and meaningful information for both trainees and supervisors. Detailed feedback on assessment performance allows registrars to improve their practice, whereas the information from pass/fail types of feedback is limited (Shute, 2008).

This notion also aligns with a programmatic approach to assessment, which treats assessment as a whole program, rather than as isolated or ad hoc assessment instruments (Schuwirth & van der Vleuten, 2011; Prideaux, 2016). The strength of an assessment program is thus judged by the program as a whole rather than by the properties of single instruments. A variety of well-chosen tools can be used where appropriate, thus providing adequate sampling across the breadth and depth of the curriculum. A key principle in this approach is that individual assessments can provide targeted feedback for learning, while high-stakes decisions can be made based on aggregating the results of multiple assessments (van der Vleuten et al., 2012). The strength of this approach is that there are multiple opportunities for gaps in learning to be identified, and remedial work to be instigated. That is, there should be no 'surprise' if a candidate fails one high-stakes instrument at the completion of training, as there are progressive measures of competency attainment throughout the training program.

Aligned and fit-for-purpose assessment

The program outcomes, curriculum content, teaching and learning strategies, and assessment and examination processes all need to be aligned (Biggs, 2003). This alignment assures that content in the curriculum is linked to the stated learning outcomes of the program. If learning is a key focus of an assessment program, it is even more crucial that the assessments align to desired program outcomes. This is particularly important as it is a well-accepted adage that assessment drives learning. The process of blueprinting is a key tool to achieve constructive alignment (Cumyn & Harris, 2012).

An assessment is fit-for-purpose when the methods used to assess a particular learning outcome are relevant, appropriate and focused on measuring the achievement of that outcome (van der Vleuten et al., 2012). There are many aspects to assessment quality. There is the often ambiguous notion of

assessment validity, which can refer to many things and comes in many forms (construct, content, face, predictive, and so on). Today, validity is typically considered in terms of evidentiary frameworks. Validity is established through arguments based on substantive evidence. Another facet to assessment quality is the criterion of psychometric robustness, defined as the quality of the psychometric properties of an assessment. The reliability of an examination is the degree to which an assessment instrument produces stable and consistent results. Examination scores can never be completely reliable; however, in a high-quality assessment, the contribution of measurement error to candidate scores should be low. Fitness-for-purpose is a functional definition of assessment quality (Schuwirth & van der Vleuten, 2011). In a program of assessment, there will be a variety of different assessment methods deployed, each designed to consider issues such as validity, reliability, feasibility and acceptability. Essentially, consideration should be given to whether each selected assessment type adequately assesses the intended relevant content.

Collaborative construction of standards

Standards can be defined as the 'degree of academic achievement established by authority, custom, or consensus and used as a fixed reference point for reporting a student's level of attainment' (Sadler, 2013: 15). Determining performance standards should be a constructive and collaborative process. Having well-articulated performance standards helps improve the transparency of assessment and student learning (Hendry, Armstrong, & Bromberger, 2012). As assessment standards are socially constructed, assessors need to engage in dialogue about standards as part of a broader understanding of the process of assessment (HEA, 2016). Further, this cannot be a 'one-off' event, as the expectations for appropriate standards evolve over time.

Defensibility of professional judgements

Professional judgement is an intrinsic element of assessment. Confidence in such judgements relies on developing appropriate standards for performance and regularly reviewing these standards. Professional judgements should be continuously reviewed through collaborative, critical reflection to ensure that they are reliable, consistent, transparent, fair, and ultimately defensible (Bloxham, den-Outer et al., 2016). Moderation techniques and forums for deliberating on performance have been shown to improve the assessment literacy of assessors and their ability to apply standards consistently and correctly (Bloxham, Hughes et al., 2016).

Fairness and consistency

A program of assessment must be defensible (in terms of the assessments being valid, reliable, robust, etc.) and transparent (which means that fair and transparent processes are used in making decisions about candidates' ability to practice). Fairness relates to the degree to which the assessment provides an equivalent measure of candidate ability independent of the personal characteristics of the candidate that are not relevant to the assessment. At the item level, this includes issues around potential bias in assessment. At the process level, fairness relates to the consistency of candidate experience and evaluation in the assessment process.

Limits and feasibility of assessments

Assessment can only do so much, and complete precision can never be fully attained. In essence, the process of assessment should aim at optimisation. Finally, it is important that changes to assessments and assessment processes are feasible within a particular context. Different health professions have different needs, foci and idiosyncrasies. Assessment processes need to embrace this heterogeneity and find ways of incorporating best practice in unique contexts.

FINDINGS: WRITTEN EXAMINATIONS

MCQ EXAMINATION

Purpose and format

The MCQ examination is a 190-minute, computer-based Multiple-Choice test that candidates undertake at a Pearson VUE testing centre. The exam is offered twice a year. RANZCP recommends that trainees attempt the examination during Stage 2 of training (and complete it by the end of Stage 2), but candidates may apply to complete this assessment after they have completed six months of FTE training. Successful completion of the examination is, however, not a prerequisite for starting Stage 3 of training.

The examination is designed to assess foundational knowledge in psychiatry covered through the Stage 1 and 2 syllabus and is set at the level expected at the end of Stage 3. The examination comprises 140 Extended Matching Questions (EMQs)¹, each worth one mark, and two Critical Analysis Problems (CAPs) worth 40 marks in total. The EMQs comprise a short stem and a number of response options. Candidates are directed to choose the single most closely associated response through directions such as selecting the most appropriate or most likely response. The number of response options varies across questions and was up to 13 possible responses for the August 2019 MCQ. The CAPs comprise a series of MCQs related to an abstract and selected tables and figures from recent research articles.

The purpose was well understood by all relevant stakeholders as a test of basic knowledge. The content is quite theoretical and knowledge-based. As assessment drives learning, it is clear that this examination drives trainees' knowledge acquisition early in training. However, there was some concern that although the test is supposed to be aimed at a junior consultant level, this is not the level of competency that trainees have actually attained when they are sitting the exam.

Item quality

Part A: EMQs

This review of the MCQ examination takes as its reference point the writing guide titled *Constructing Written Test Questions for the Basic and Clinical Sciences* (National Board of Medical Examiners, 4th edition, 2016). This work has been influential in guiding and reforming MCQ writing practices in medical education, for both undergraduate and postgraduate contexts, since its first publication in 1996. The MCQ format is widely used in medical education to assess factual recall, application of knowledge and clinical reasoning. In other fields, including the biosciences, the MCQ format is often regarded as being restricted to assessing factual recall. However, the NBME guide has been instrumental in arguing for and exemplifying the use of the MCQ format for assessing higher order reasoning skills, including clinical-based reasoning and judgement.

The NBME guide currently endorses only two types of MCQs for both basic and clinical sciences: the 'single best answer' (SBA) format, and the 'extended matching format' (EMQ). As noted below, the RANZCP MCQ examination reflects a somewhat idiosyncratic mix of these formats. On the whole, the quality of items in the MCQ exam is relatively high, in terms of offering authentic tasks and contexts to assess the relevant knowledge, application, and reasoning of trainees, with few obvious and irrelevant cues to enable respondents to select the key using 'testwiseness' strategies (NBME, 2016).

¹ We have referred to the form of questions as EMQs as they are described on the College website; however, we recognise that these questions are now in transition because the EMQ format cannot be supported by a computer-based delivery of questions. This is discussed later in the report.

Question format

The items are labelled 'EMQs' in the test itself. To be consistent with use in the wider medical education literature, an EMQ should consist of a number of scenarios/questions linked to a list of options corresponding to an overarching theme, usually with the aim of assessing clinical reasoning. Given the presence of common options across multiple questions (e.g. EMQs 90–92), it appears that these items originally formed part of conventional EMQ sets, and have since been 'broken up' to form individual questions. It is our understanding that this occurred with the transition to online delivery of the exam, with the need to display one question per page.

Although there is a sound pragmatic reason for this practice, an unintended effect is the presence of inappropriate or implausible options for many questions (see below under distractor quality). This is a result of breaking up EMQ sets, whose options are developed to represent a wide range of possibilities (diagnoses, investigations, management options, etc.) specifically for numerous related but different presentations. This means that while an option may be largely irrelevant or implausible for one scenario, it should come into play for one or more others, so that each distractor may be a plausible response to an overall theme. Many options lose this relevance when used as a stand-alone question, resulting in a relatively large number of irrelevant and non-functional distractors. This can be confirmed from the item analysis, which shows many items having only few selected options (e.g. EMQs 13, 18, 35, 40, 44, and 45).

The current practice of using broken-up EMQs is therefore likely to affect the authenticity, level of difficulty, and ultimately the quality of the MCQ exam. We recognise that work in this area is occurring progressively. If it is not possible to re-aggregate the EMQs as originally written, we recommend that the College continue to focus on editing each item to have a more coherent and plausible set of four or five options. While the research suggests that three options in total are usually sufficient (Rodriguez, 2005), four options are usually recommended for reasons of symmetry. Five options are common in medical education, so may be preferred for contextual reasons, but the addition of the fifth option usually offers little extra value. The number of options need not necessarily be the same for each item (see Zoanetti et al., 2013), although in practice most organisations use a consistent number of questions for ease of development and delivery.

Authenticity of questions

As noted above, the items largely offer apparently genuine and well-contextualised presentation to appropriately assess the clinical reasoning of candidates. However, several questions present minimal context and are little more than knowledge or definition matching questions (e.g. EMQs 4, 9, 10, 19, and 22–24). Such questions should be reviewed to confirm that they appropriately address the relevant learning outcomes or competencies.

Several questions that do address clinical knowledge nevertheless have minimally elaborated contexts. An example of this is EMQ1 which seeks an explanation for the 'emergence of tardive dyskinesia in a patient who has taken haloperidol for three years'. These types of questions not only seem less authentic, lacking the patient factors, comorbidities, or assessment findings commonly included in most of the other questions on the exam, but they may arguably encourage precipitous or superficial conclusions rather than multifactorial clinical reasoning. Again, this is a substantive (rather than technical) matter that should be reviewed by the appropriate subject matter experts (SMEs).

Technical quality of questions

Several MCQ items appear flawed on technical grounds, with respect to commonly accepted drafting principles. The following are examples noted in a single exam (August 2019), and as such may or may not reflect systematic flaws in the writing process.

- Further to the above comments about irrelevant options, many questions contain 'heterogeneous' options that do not appear to form a coherent set of options for a focused task (e.g. EMQ 12, 27).
- Some options are written inconsistently in relation to the lead-in. For example, EMQ1 which calls for an explanation, so options should not appear to be an independent statement of fact (c.f. B and C).
- The authenticity and/or effectiveness of questions that ask for the 'BEST MATCHING phenomenon' may be queried. Is the appropriate criterion for such judgements clear to candidates? (This may help explain the low discrimination indices observed in EMQs 30 and 103.)
- There are inconsistencies in the level of patient detail across the exam.
- Several distractors seem highly implausible and able to be rejected by untrained lay readers (e.g. EMQ17, social skills training; EMQ27, Immanuel Kant; and EMQ33, fetishism/ frotteurism/ transvestitism). As discussed above, this issue is likely to be a result of the stand-alone use of sets of EMQs.

The above-mentioned flaws in items are by far in the minority; as noted above, most questions on the exam provide an authentic clinical task, a focused lead-in, and a coherent set of plausible options. Further potentially problematic questions may also be identifiable from the item analysis, and it is noted that the provided analysis has already identified such questions for review.

Part B: CAPs

These problems, based on published research abstracts, appear to present an authentic and meaningful task for candidates through the MCQ (SBA) format. Most questions provide relevant and reasoning-level questions, appropriate to the nature of the task. Other questions, however, remain factual in nature, and not dependant at all on critical appraisal of the exam, e.g. (from the August 2019 exam) CAP1 Qs 2–4, CAP2 Qs 1–3. This may be intentional to provide easier 'lead-up' questions; however, some of the questions appear to be answerable through little more than language comprehension (e.g. Q1.5 – match between 'adjustment' in lead-in and 'correction' in key; Q2.6 utility/effectiveness).

The table-based questions in CAP1, on the other hand, are consistently focused on the provided material and demand higher-level understanding and reasoning. Presumably the data is more easily readable than appears to be the case on-screen.

Some of the questions use a cumbersome or fill-in-the-blank format (e.g. Q1.5, Q2.9–10) that is less authentic than the standard SBA format.

Constructive alignment

The written examinations play a major role in sampling content from across the curriculum. There was general acknowledgment that although the examinations aligned to the curriculum, there was probably not complete coverage.

An assessment blueprint outlines a plan or a map, usually in the form of a grid or a matrix, to guide item construction and selection for assessments in a way that is planned and replicable (Hamdy, 2006). The blueprint should systematically describe the assessment methods used to assess relevant learning outcomes and provide a template to describe the content of specific exams at the item level. Planning assessment against a blueprint ensures that there is appropriate coverage of the stated learning objectives or expected competencies and that the assessments provide a representative sample of performance (General Medical Council, 2017). Appropriate blueprinting of assessments minimises important risks to validity. Difficulties with construct representation can result when sampling is not

representative but is biased in some way (e.g. weighting which does not reflect the coverage in the course content) (Ray et al., 2018). An appropriately constructed blueprint minimises this threat by ensuring that weighting of items reflects coverage of the content to be assessed.

Blueprinting is a process that is part of a broader program assessment framework. Unlike curriculum frameworks which typically outline program learning outcomes, assessment frameworks specify which learning outcomes are to be assessed and how they will be assessed (Jago, 2009; Pearce et al., 2015). A well-designed assessment framework provides a reference point to determine whether assessment tasks are appropriate measures of particular learning outcomes and whether there is appropriate coverage of the learning outcomes overall (Pearce et al., 2015). Test-specific blueprinting arises from this framework to specify what aspects will be addressed in a given assessment. Assessment frameworks and blueprinting are important components to fulfil the educational principle of constructive alignment. While blueprinting processes exist for the RANZCP written examinations, no assessment framework was sighted.

The MCQ Examination was blueprinted according to the curriculum at the item level. This allows for detailed analysis of candidate performance against specific topics. The items are also aggregated into assessment domains from the syllabus (e.g. 'ethics', 'professional communication and liaison', specific disorders, specific areas of practice and others). Over time, the blueprinting process has become more refined and appears, in its current form, to be thorough, considered and robust. The detail at the item level is particularly notable. The Written Examinations subcommittee can track the number of items administered over time in the different topic areas and domains. They can review item selection across these areas and identify gaps or underrepresented areas for future examination development. The blueprinting enhances the reporting options to candidates in their feedback letters. Trainees are aware of the blueprint and the relative importance placed on particular topics in the syllabus.

Fitness-for-purpose

As an exam that assesses fundamental background knowledge, the general consensus was that the MCQ examination was fit-for-purpose. The exam was seen as important in encouraging trainees to ensure their foundational knowledge was sound. There were some concerns that the items were too easy, 'dumbed down', and could be passed using test-wise strategies. Most stakeholders agreed that the examination was not well targeted to an appropriate standard, in part due to the fact that trainees can sit the examination as early as after 6 months of FTE training. The purported standard (that of a junior consultant) requires rethinking as candidates have typically only completed 12 months of training. This means that the items typically focus on foundational knowledge at the level of a senior registrar. The focus could be honed to the fundamentals of psychiatric training and the examination could become a hurdle to training progression. The EMQ format was seen to be suboptimal, and committee members were interested in exploring options for item reform.

Fourteen members of the Committee for Exams responded to questions regarding the quality of the MCQ examinations in the survey. On a scale from 0 to 100, the mean ratings were as follows: quality of the MCQ exam content (59.3), validity of the MCQ exam format (62.5) and usefulness of the MCQ exam score (62.8).

Trainees and Specialist International Medical Graduates (SIMGs) believed that they had adequate time to prepare for the exam (88% agreed); however, responses on most other items were variable with similar proportions of candidates agreeing and disagreeing on several items. For instance, 53% of respondents agreed that the curriculum clearly identifies what needs to be learnt for examination success, 67% agreed that they had a good understanding of what is assessed in the exam, and 48% agreed that the skills assessed in the exam are well aligned with their training program. Many respondents agreed that the MCQ examination was difficult (76%), but most felt that they spent enough

time preparing (91%), were well prepared for the format of the MCQ exam (89%), and that their skill level was right for the exam (93%).

Directors of training and supervisors agreed that they had a good understanding of what was assessed in the MCQ examination (79%). The majority of respondents also agreed that the skills assessed in the MCQ exam are well aligned with the training program (79%), that the format of the MCQ exam is appropriate for the intended trait being assessed (81%), and that it is important to assess the traits measured in the MCQ examination (91%). Most respondents believed that they had a good sense of trainee skill level prior to them attempting the MCQ exam (73% agreed), and the majority were confident that most of their trainees who attempted the MCQ exam were progressed enough to pass it (76% agreed). In general, fewer respondents found the results of the MCQ examination useful for identifying training needs. Only 26% agreed that they identified a trainee's skill level from their MCQ exam results, 29% agreed that they identified a trainee's strengths and weaknesses from their MCQ exam results, 51% agreed that they used the results to identify the area of support required, 40% agreed that they used the MCQ exam results to identify training resources needed, and only 35% used the MCQ exam results to evaluate their training programs.

ESSAY-STYLE EXAMINATION

Purpose and format

The Essay-style examination is a 180-minute, paper-based examination that candidates undertake at a testing centre with an additional 10 minutes of reading only time. The exam is offered twice a year. RANZCP recommends that trainees attempt the examination during Stage 3 of training, but candidates may apply to complete this assessment after they have completed 18 months of FTE training, with at least six months in Stage 2. Successful completion of the MCQ examination is also not a prerequisite for attempting the Essay-style exam.

The examination is clinically focused and is designed to assess critical thinking in relation to clinical practice and broader issues relevant to the psychiatric profession, including sociocultural and ethical issues, as well as the application of clinical knowledge. The examination comprises one Critical Essay Question (CEQ) worth 40 marks and 4–6 Modified Essay Questions (MEQs) worth up to 30 marks each.

This exam assesses more than basic knowledge. Trainees must demonstrate a higher level of understanding and show that they can apply their psychiatric knowledge in context. The CEQ requires candidates to write clearly and concisely under time pressure and demonstrate that they can understand and analyse complex ethical, societal and historical issues relevant to psychiatry. Candidates are required to pass the CEQ component of the Essay-style examination in order to pass overall.

Item quality and rubric design

Part A: MEQs

The MEQ format represents a relatively common approach to constructed-response (CR) questions in medical education, enabling the assessment of generative reasoning in clinical contexts. Constructed-response questions are evaluated according to three key elements: the task itself (including questions to candidates), a marking guide (or rubric), and a scoring sheet. The nature of the tasks, both in the information provided and the questions to the candidates (judging by the August 2019 MEQs), appear to utilise the CR-format appropriately by asking candidates to do more than simply respond with factual information; candidates are required to describe, explain, justify and/or discuss their proposed actions or reasoning in relation to the provided clinical scenarios. Such questions can be described as 'semi-constrained' (Ahmed & Pollitt, 2011), in the sense that candidates are directed to responding to specific questions but there remains a certain degree of freedom in the nature of the responses which can be

rewarded. The key verbs are helpfully glossed using consistent terminology, which candidates have presumably been introduced to prior to the examination.

For the RANZCP MEQ exam, the marking guide is presented together with the scoring sheet. On the whole, the marking guides and score sheets are clearly set out, using an efficient design to efficiently record (and subsequently process) marker scores. There is a useful note to markers, anticipating potential confusion around the potential discrepancy between 'bubble scores' and the total marks for the question. In some cases, there are further content-based notes to markers to support judgement of candidate responses (e.g. MEQ 3D).

The marking guides are well-structured, separating and outlining different components to be judged and scored for each sub-question. This represents an analytical approach to marking, rather than holistic (i.e. an overall qualitative judgement), and this appears to align well with the relatively focused nature of the MEQ sub-questions. The MEQ marking guides are based on a category or levels approach, whereby markers judge or 'code' the overall quality of the response in relation to each criterion, as opposed to points-based marking that simply assigns a mark (or part thereof) for every correct and relevant point made by candidates (in accordance with the guide). Levels-based approaches are highly appropriate for reasoning-based constructs, and likely to be more effective and discriminate better amongst responses than points-based approaches in this context (Ahmed & Pollitt, 2011). However, some of the marking guides appear somewhat underdeveloped, from the perspective of providing sufficient guidance to markers. An example is MEQ 4.3, which invites candidates to 'discuss' possible interventions. The reproduction of the term 'discuss' in each element (A–D) does not seem to offer much assistance or clarity to markers, nor does the suggested scoring illustrate how the 'debate' component of 'discuss' should be identified and scored. Furthermore, explicit criteria for each level, including which score or code to assign to a response, is missing for most of the questions. Consequently, there appears to be considerable room for variation in judgement between markers without further guidance as to whether a response in one criterion or component should be categorised as, say, a 0, 1, 2 or 3.

The academic literature on marking schemes often differentiates between different 'levels' of guidance for markers (Ahmed & Pollitt, 2011), as follows:

- Level 3: Rule or principle to differentiate responses against a specific quality or criterion
- Level 2: List of good responses vs poor responses (with examples)
- Level 1: List of good or acceptable responses (with examples)
- Level 0: Only a 'model answer' or no guidance at all

According to this taxonomy, most of the marking guides used in the MEQs belong to level 1. While more useful than the 'model answer' type commonly used in medical and bioscience education, they do not currently contain information relating to poor or unacceptable answers (level 2) or the underlying rule or principle to enable grading of unanticipated or unspecified candidate responses. Including such information for markers (where appropriate) would likely improve the reliability, discrimination and effectiveness of the marking process (Ahmed & Pollitt, 2011; Pinot de Moira, 2013).

Part B: CEQs

The CEQ is a variation of the CR format, a relatively unstructured format enabling the assessment of reasoning in clinical contexts. Its most common form is the 'essay', which is explicitly the format used in the RANZCP exam, where the task is to 'critically discuss' a supplied quotation. Notably in this instance, the key verb phrase 'critically discuss' is not glossed, although this may be suitably addressed during training and/or exam preparation.

The CEQ marking guide is set out in the same format and structured analytically as for the MEQ guides. The marking guide provides references to relevant Fellowship Competencies and includes the relevant

weightings; again, these are appropriately made available to candidates. The analytical structure and description of the various levels within each competency is commendable.

The marking guide taxonomy is slightly modified for such 'unconstrained questions' as the CEQ (Ahmed & Pollitt, 2011), in recognition of the different requirements of the task:

- Level 3: Rule or principle *specific to the task*, which differentiate responses against criteria (i.e. a 'specific trait interpretation')
- Level 2: Analytical structure with generic descriptions (i.e. applicable to different tasks)
- Level 1: Holistic structure (i.e. overall judgement only, not against different criteria)
- Level 0: 'Model answer' or no guidance at all

In this case, the CEQ marking guide fits with level 2, since the criteria appear to be 'generic' in nature. Therefore, the characteristics and level of candidates' responses which specifically address the key elements of the task is left to markers to work out and decide how to apply and score. However, as noted above in relation to the MEQs, the current structure of the CEQ marking guide represents a relatively high level of marking guide explicitness, especially in relation to other common practices in medical and bioscience education. While the addition of a 'specific trait interpretation' element would further improve the quality of the marking guide, it is recognised that there may be pragmatic reasons that dictate a preference for the more generic version of the marking scheme.

Overall utility of the marking guides

There were suggestions that, in practice, examiners find the marking guides difficult to use. Some worried that the MEQ marking guides were 'overdeveloped' resulting in quite rigid grids that don't allow for markers to execute their judgment and discretion. Conversely, some worried that the CEQ marking guides were too generic and missing example responses. The challenge with marking guides is in providing sufficient descriptions for markers to assess a piece of work in a way that was intended in the question and consistent in its execution, while still providing sufficient flexibility to code the quality of a response into different levels of performance. This can be optimised by developing robust principles-based rubrics that are task-specific.

Constructive alignment

As mentioned earlier, the written examinations play a major role in sampling content from across the curriculum. There was general acknowledgment that although the examinations aligned to the curriculum, there was probably not complete coverage. The smaller number of items and more in-depth focus on testing higher-order thinking in the Essay-style examination places limits on the breadth of sampling. Although some gaps in specific topic areas for any one examination instantiation are necessary, many examiners thought they could do a better job of assessing some of the 'harder to assess' domains.

A test-specific blueprint was sighted for the Essay-style examination. Blueprinting to specific competencies at the item level allows for a detailed analysis of candidate performance against specific topics. The detail at the item level is particularly notable as the Written Examinations subcommittee can track the number of items administered over time against specific competencies. The blueprinting enhances the reporting options to candidates in their feedback letters.

As mentioned in the MCQ Examination section above, blueprinting is a process that is part of a broader program assessment framework. Although blueprinting processes exist for the written examinations, no assessment framework was sighted. Although items are mapped to competencies, there was no framework that provided guidelines to examiners around appropriate curriculum coverage in any one administration. Appropriate blueprinting should begin with a clear definition of the purpose of the assessment (usually articulated in the assessment framework), to ensure that the development process

produces assessments that are matched to the expectations of candidates at particular levels (Hamdy, 2006; Sales, Sturrock, Boursicot, & Dacre, 2010). Blueprinting involves specifying the competencies to be assessed in terms of content (topics or subjects) or processes (procedural skills) or a combination of both (Raymond & Grande, 2019).

Thus, aside from mapping just to content, other categories of information can be captured in the cells of the blueprint. Information at the test-specific level might also capture information on other dimensions (for example, difficulty or anticipated difficulty of items, cognitive processes). This would allow for a more explicit identification of the skills being assessed in the MEQs and CEQs, and how they link back to the training program. The item-specific information may also show how the item in a blueprint cell maps to a specific assessment format or to information about assessment objectives or competencies outlined in an assessment framework.

Fitness-for-purpose

The general consensus was that the Essay-style examination was important in assessing whether trainees could take a broad view to demonstrate their awareness of social and cultural issues beyond the medical model. It assessed their knowledge on the history and philosophy of evidence-based practice, their capacity to construct a reasoned written argument that was rational and respectful of a range of views, and to bring evidence to bear on a topic. However, the lack of an assessment framework that clearly identifies the link between what is being assessed in the Essay-style examination and the learning outcomes of the broader training program is concerning. Specifically, it is unclear how the skills being assessed under the examination conditions in the CEQ reflects the real-world practice of being a psychiatrist.

There was strong agreement that the MEQs assessed integrated aspects of training and many saw the advantage of the format; however, the MEQ marking (with subsections marked by different examiners) did not provide a clear overall perspective on candidates' skills. Some worried that able candidates would struggle to demonstrate their proficiency due to time constraints.

Some were concerned that the CEQs were somewhat straightforward and that with appropriate practice and examination technique, passing was easy. The rationale for the weighting of marks (40 for CEQs and 180 for MEQs) was queried by many of those interviewed. A conjunctive rule is set whereby candidates must pass the CEQs to pass overall (because in the past when this rule did not apply the standard of response on the CEQ was exceptionally poor), but it was unclear why the CEQs and MEQs were aggregated at all. They appeared to be functioning as two separate assessments.

The low pass rates on the Essay-style examination were a concern to many. There was also an acknowledgement that this examination was more difficult for trainees with non-English speaking backgrounds. Indeed, evidence from the college-administered survey indicated that there were far more candidates than expected in the unsuccessful category (first attempt on the Essay-style Examination) from the group that did not have English as their first language.

Thirteen members of the Committee for Exams responded to questions regarding the quality of the Essay-style examinations in the survey. On a scale from 0 to 100, the mean ratings were as follows: quality of the Essay-style exam content (68.8), validity of the Essay-style exam format (60.8) and usefulness of the Essay-style exam score (54.5).

Trainees and SIMGs were generally less favourable for the Essay-style exam than for the MCQ exam. Respondents generally agreed that they had adequate time to prepare for the Essay-style exam (67%); however, few respondents agreed that the curriculum clearly identifies what needs to be learnt for examination success (24%). There were also fewer respondents who agreed that they had a good understanding of what is assessed in the exam (34%), or agreed that the skills assessed in the exam

were well aligned with their training program (29%). Most respondents agreed that the Essay-style examination was difficult (94%), but most felt that they spent enough time preparing (89%), were well prepared for the format of the Essay-style exam (76%), and that their skill level was right for the exam (82%).

The majority of Directors of training and supervisors believed that they had a good understanding of what was assessed in the Essay-style examination (73% agreed). Respondents also tended to agree that it was important to assess the traits measured in the Essay-style exam (79% agreed), but were less inclined to agree that the skills assessed in the Essay-style exam were well aligned with the training program (63%) or that the format of the Essay-style exam was appropriate for the intended trait being assessed (61%). Most respondents agreed that they had a good sense of trainee skill level prior to them attempting the Essay-style exam (76%), but fewer were confident that most of their trainees who attempted the Essay-style exam were progressed enough to pass it (56% agreed). In general, fewer respondents found the results of the Essay-style examination useful for identifying training needs. Only 28% agreed that they identified a trainee's skill level from their Essay-style exam results, 35% agreed that they identified a trainee's strengths and weaknesses from their Essay-style exam results, 43% agreed that they used the results to identify the area of support required, 38% agreed that they used the Essay-style exam results to identify training resources needed, and only 30% used the Essay-style exam results to evaluate their training programs.

FINDINGS: WRITTEN EXAMINATION PROCESSES

ITEM DEVELOPMENT

The Written Examinations subcommittee follows quite a rigorous process for drafting, selecting and reviewing items. As with all item construction, there is always room for quality improvement and enhancement. The general feeling was that the quality of the processes for development and review have improved over time. Although some of the processes have evolved organically and in a piecemeal fashion over time, overall the processes appear robust.

There is an existing MCQ database with a large number of items, although there was acknowledgement that there were some problems with the structure of the database and the usefulness of some of the items in there due to their structure, relevance and quality. There is also a small bank of questions for the Essay-style examination. A contingency exam is kept aside as a 'back up' in case something goes wrong during delivery. A small number of questions from time to time are re-used in order to compare differences in cohorts, but the overall focus is often on developing new items.

In line with best practice, the blueprint appears to be the starting point for question development. MCQ and MEQ writing workshops are run face-to-face annually, usually over a weekend. The process relies on a high-level of engagement from interested committee members. It appears that the consultants who are involved in the question development processes are engaged and committed. The College is fortunate to have such committed members. They strive to ensure that the questions are grounded in contemporary knowledge and practice, and are appropriately referenced. Any examiner on the Written subcommittee can find and submit a suggested quote for the CEQ. Committee members subsequently review all suggestions and select the quote for each exam.

Some training on item writing is provided but in more of an ad hoc fashion. The participants develop questions and extensive peer review of questions occurs at this workshop. The focus is often on writing new questions for the areas in the blueprint that have lower numbers. After the workshop, participants submit questions throughout the year. These are reviewed by the chairs, who may make changes and remove ambiguities in an iterative process over the year. Developing items in a collaborative and

iterative fashion is good practice. The MCQs and MEQs are reviewed multiple times by committee members who focus on these specific formats. The reviews ensure that the wordings and marking guides are clear. In order to evaluate the quality of the marking guide, committee members also attempt to answer the questions in the appropriate time allocated in the examination to judge whether the expectations in the marking guides are appropriate, which is good practice.

EXAM DELIVERY

The MCQ examination is computed-based and is delivered through Pearson VUE testing centres. There were some concerns that limitations with the platform have led to inauthentic presentations of content, due to inability to cope with the EMQ item type. When incidents occur in the delivery of the examination, incident reports are provided to the College.

The Essay-style examination is delivered by paper. This makes marking difficult, as markers have to decipher candidate handwriting. With the large number of candidates involved in any one examination sitting, the process (while thorough) appears inefficient on a number of levels. Although the possibility of delivering the exam online has been explored previously, it would be worth further exploring the options of moving to an online platform to deliver this examination.

MARKING PROCESSES

In line with best practice, the MEQ and CEQ tasks are double marked. On review of the available data, it was evident that marking discordances are resolved by taking the average of the two marks when below a certain discordance threshold. This appeared to be set at 25%. It would be worth considering lowering this threshold, especially when reviewing results from borderline candidates.

Several committee members noted that it can be difficult to find enough people to mark the Essay-style examination. There were concerns that the validity and reliability of the marking may be compromised with a wider group of people. With appropriate training and calibration, this can be mitigated against. It was also suggested that the marking sheets should be imported into an online platform. At present, it is possible for markers to leave sections of the rubric incomplete, which results in significant efforts to determine the missing marks. Indeed, electronic marking removes the potential of many human errors and enhances the efficiency of the marking process.

PSYCHOMETRIC ANALYSES

A review of all submitted examination data was conducted by ACER psychometricians. For the MCQ examination, we relied entirely upon the supplied materials. For the Essay-style examination data, we were additionally able to draw upon our involvement with data analysis and checking during April and May in 2019, including our participation in the standard setting committee meetings for these assessments.

Item analysis

The comprehensive item analysis information that was evident in the supplied files was encouraging. The augmentation or replacement of the previous classical test theory statistics (and some simplistic thresholds) with Rasch test and item statistics will provide more comprehensive information for scrutinising the fitness-for-purpose and quality of the exam and individual items. It appeared that the materials were still in working draft form, and so we assume they have not yet found their final structure and format.

In light of this, it will be possible to standardise and rationalise the item analysis output report and to enhance its aesthetic appeal, a process which is no doubt already under way. This includes the possibility of automating its production complete with conditional highlighting and automated population

of pre-set tables, graphs and in-text metrics, for example using the R statistical programming environment. RANZCP might consider whether developing a set of templates in this way would represent a worthwhile investment. Our experience is that this is beneficial for programs that require consistent reports on a predictable operational schedule with tight timeframes, though for a single exam only, the investment of effort still needs to be weighed up.

Item review

In addition to the various metrics presented in the working draft report, Item Characteristic Curves should also be inspected. This is crucial in the context of high-stakes exams, particularly those operating in contexts where candidates might have received their training and prior education at a variety of institutions with potentially conflicting interpretations of acceptable answers to exam questions. While the item review processes are supposed to prevent such items from appearing in the exam, it is not always possible to anticipate what an acceptable alternative interpretation of a question might be. The curves are the best way to detect if high-ability candidates, even a small proportion, have opted for one of the distractors at levels not predicted by the measurement model. We therefore encourage a careful review of category characteristic curves, particularly for items that 'only just' pass through a check of the usual item statistics, such as facility, discrimination and fit.

Test targeting

With respect to test targeting, reference to the Rasch variable map was a strong inclusion. If the location of the pass standard from the standard setting process was available, then a horizontal line or discussion about the location of the cut score relative to the density of the items could be of interest. This reflects that test development efforts ought to provide sufficient and ideally very concentrated targeting of items around the intended pass standard.

Differential Item Functioning

We observed that Differential Item Functioning (DIF) analysis was potentially under consideration, as indicated by a placeholder in one of the reports. Given the small cohort sizes, this may not result in a particularly reliable analysis, though it could still be instructive for detecting significant outliers.

Data management

Improvements to the format and storage and processing of assessment data are still possible. On Page 8 of the Standard Operating Procedures document, manually updating scores following a change to a scoring key should be avoided. Instead, we would recommend updating the master key, and re-running the scoring of items. This ensures that no candidate's score will be incorrectly updated and that a more manageable number of changes (i.e. to the key only) are required. If this cannot be done, then we would still warn against manually over-riding item scores. Instead, we would recommend the addition of two new columns, one for the updated item scores, and one to run a systematic validation check on the updates. While the exact details of these processes are not known, and our concerns could be misguided, we simply want to rule out that a risky approach is in place.

Weighting of items

In general, the need to weight individual questions is questionable, though we appreciate that policy and perception can play a role. We argue that weighting of the relative coverage of certain knowledge and skills ought to be addressed through the blueprinting entirely, and less so by arbitrary weighting of certain question types. If these questions are intended to probe deeper knowledge, but with only one mark available, perhaps they would better be reformulated as MEQs and assessed in the Essay-style examination.

Linking and equating

Although there was an indication that a small number of items are repeated across administrations, there was no evidence that psychometric linking and test equating practices are occurring. It is likely that adding these would be of uncertain value as item exposure is a risk when linking examinations and the standard setting approaches ensure that appropriate cut scores are set for each examination.

Perspectives from stakeholders

Comprehensive item statistics are provided to the committee for review. The item analysis flags items for review. However, it appeared that there was often limited time to go through the statistics in detail. Many panel members noted that they had a limited understanding of the analyses and just had to have a 'level of faith' about the process. Some felt uncomfortable knowing that they did not fully understand the analysis process or how to properly review the psychometric properties of items. Although the analyses are robust, the data should be used as a pragmatic tool by the committee in their deliberations and decision-making. As many are not confident on how to interpret these data, some targeted training for committee members would be worthwhile.

STANDARD SETTING AND CUT-SCORE DETERMINATION

Standard setting is a process of systematically determining the score on an examination that represents the minimum acceptable level of performance. Standard setting is a more defensible approach than determining a pass mark (or level of performance) by default, custom or arbitrarily. Essentially, the standard setting process attempts to define what level of performance on an exam is to be regarded as 'just good enough' for the current stage of a learning programme or for certification of a defined level of competence. There are many different methods for standard setting, all of which aim to collate expert judgement in a defensible and valid way. A useful distinction can be made between *item-based* or *performance-based* methods. Item-based methods use the predicted performance of minimally competent candidates to determine the appropriate cut score; this method is particularly appropriate for written exams of underlying knowledge and understanding. Performance-based methods rely on actual performance in the exam, whereby data on the performance of minimally competent candidates is subsequently used to calculate an appropriate cut score.

The Committee for Examinations should be commended for producing a (confidential) document relating to the 'Written Examination Standard Setting Process'. This document clearly outlines the item-based methods employed in standard setting the Written examinations. In line with best practice, it emphasises that the methods used are criterion-referenced rather than norm-referenced. It was also particularly pleasing to see an articulation of what a minimally competent candidate (MCC) looks like for this specific context, and to underscore that a MCC is different to an 'average' or 'aspirational candidate'. The standard setting usually occurs at a face-to-face meeting. The facilitation of 'satellite' standard setting is also good practice, and ensures that perspectives on the minimal acceptable passing standard from outside the committee is taken into account. Nevertheless, it is important to ensure consistent training around conceptions of the MCC for these groups, and to consider how to resolve large differences between the judgments of the Committee and the Satellite should they arise.

The MCQ examination (both EMQs and CAPs) standards are set using a modified Ebel method. The Essay-style examination (both MEQs and CEQs) standards are set using a variation on the extended Angoff method (although the document does not identify that this is the method being used). While both of these criterion-based methods are appropriate for this context, the efficacy of the precise implementation does raise certain questions regarding whether these methods are fit-for-purpose in their current form.

Perspectives from stakeholders

As already flagged earlier in this report, there were many concerns that the standard for passing exams is meant to be at the junior consultant level, even though there is a necessary mismatch between this standard and the level of competency trainees will have reached early in training. Many were concerned that the standard was also set too low and, subsequently, the pass rates are too high.

There was strong agreement that the Ebel method for the MCQs was highly convoluted. Indeed, it seems that the existing implementation has evolved organically over time and has become overly complex. Most did not understand how the outcomes of the method are determined. Some 'didn't understand it', and some 'didn't want to'. Some see the percentage aspect (of identifying what proportion are likely to get a question correct) as difficult to determine and ultimately flawed. Most struggle with the process. One stakeholder noted that the process 'felt rigorous', but that does not mean that it is. In a drive to be diligent and scrupulous, it may have been 'overdone'.

Options for consideration

In particular, the modified Ebel method used for the MCQ examination should be reviewed. Rather than assigning a selection of items to members (breaking into groups and doing a sub-set of the examination), each panel member should be involved in the standard setting process for the complete set of items. Making judgments relating to both relevance and difficulty should be undertaken in the blueprinting stage of development. The ratings on weightings relating to essential and non-essential elements should also be part of the process of writing and selecting items.

The Angoff method is one of the most widely used, accepted and documented item-based methods of standard setting (Norcini, 2003). The extended Angoff method works well for constructed-response items (such as in the Essay-style examination), although there are many different ways to implement the method. For the MCQ examination, the panel could review items one-by-one and estimate the probability (either as a percentage score or as a simple Yes/No judgement) that minimally competent candidates would answer each item correctly. The percentage estimates for each test item, by each judge, are then averaged, and those averages are used to determine the passing score for the exam. In the case of the Yes/No method, the number of cases deemed to be achievable by minimally competent candidates may be added up for each examiner and then averaged. Alternatively, a proportion of the Yes judgments can be aggregated to form a cut score (another variant can include a 'Maybe' judgement, which is usually counted as a half-value). Successful standard setting produces a reasonable and defensible cut score for high-stakes exams. If appropriately implemented, the Angoff method produces a cut-off score corresponding to the level of performance which the minimally competent candidate is likely to achieve on a test. It would be worth trialling simpler standard setting methods that are less time consuming, such as the Yes/No Angoff method (see Impara & Plake, 1997; Plake & Cizek, 2012) or the Bookmark Method (Lewis et al., 1996). A second round of standard setting could also be facilitated using impact data from the performance of the cohort.

Overall, we feel that the standard setting methods should be reviewed, with the current approach either simplified or replaced by a simpler method. The kinds of simplifications we would recommend apply both to the method itself, and to operational aspects like data storage and calculations. With respect to the latter, we recommend:

- use of a consistent data format (and not creating separate columns that start in a different row number from the usual structure for this information); below is an example from the file "19) MCQ Exam Result Calcs - FormulaProtect.xls"

	A	B	C	D	E	F	G	H
3	Conversion table A			Accurate score	RCS		Conversion table B	
4	Accurate score	RCS		20			Accurate score	RCS
5	30	1		20.542			20	

- the separation of calculation spreadsheets from spreadsheets that contain the raw data from judges
- the removal of any unnecessary adjustments, weightings or calculation steps where possible
- the use of standard errors (for sub-scales) to support policies about pass–fail decisions for candidates within the error width of the overall pass mark (this would simplify the deliberations required at the standard setting committee meeting and potentially mitigate against biases).

With respect to the last point, we appreciate that standard errors do appear in the provided spreadsheet. However, the way that these are calculated is unclear (as the spreadsheet did not include formulas). It is important to acknowledge that standard setting is a judgmental process and the standards obtained from that process have error associated with them. Subsequently, all pass marks established through standard setting processes should apply an error margin around the cut score. These are typically calculated using a standard error formula based on variations of rater judgment. The precise values used are ultimately judgments that should be based on College policy and precedent.

BORDERLINE CANDIDATE REVIEW

After an examination sitting, the Committee for Examinations reviews detailed results in a teleconference. The meeting papers include outcomes from the standard setting, impact analysis on the cohort, de-identified incident reports, detailed result summaries, and de-identified candidate marks. The committee reviews all borderline candidates who are close to passing and may determine that a 'conceded pass' result is appropriate in some circumstances.

Although we did not take part in one of these review meetings as part of this project, through reviewing the submitted documentation and talking through the processes with stakeholders, the process appears robust and thorough. Some notable aspects of the review process that reflect good practice include the following:

- ensuring that no candidate identifiers are presented to the committee
- reviewing specific details on cohort performance across the different examination components
- reviewing frequency distributions of candidate scores
- reviewing the impact of the standard setting outcomes on the cohort
- comparing previous cut scores and pass rates from earlier sittings.

One area identified for improvement was the criteria used to determine which candidates fall into the borderline range and thus that require further review. It was apparent that there was not a clearly defined process. Upon identifying these candidates, deliberations by the panel should focus on available data using an evidentiary framework. That is, where small deficiencies in performance in certain domains are identified, other evidence from across the assessment program could be used to inform decision-making.

TRAINING AND CALIBRATION OF EXAMINERS

While there is no formal training pathway for new examiners, informal training and induction does occur. For example, at question writing workshops, presentations are given on characteristics of high quality and poor quality items, and a basic primer on item statistics is given. It was generally acknowledged that more could be done in terms of training and support for both new and experienced examiners.

In terms of calibration, there are useful guides to marking the constructed-response items in the Essay-style examination. These provide instructions for examiners on how to assign marks and what level of detail they should expect for different questions. For new markers, they 'parallel mark' for the first time to get feedback on whether they are applying the appropriate standard and executing the marking guide as intended. Some calibration exercises occur prior to full-scale marking where random scripts are sent to co-markers and a calibration lead and they discuss marks via telephone. For the CEQs, examples of good, average and poor essays are selected and shown to the markers prior to full-scale marking. These processes seem reasonable; however, the precise details and policies around these moderation practices were unclear.

SUPPORT FOR CANDIDATES

Before undertaking their examinations, potential candidates are given access to a wide range of resources to help them prepare. Trainees have access to this material through their Learn*it* system. The amount of material made available to trainees to support them in their examination preparation is commendable. It was surprising then, that only 49% of Trainees and SIMGs agreed that there were good resources available to help develop the appropriate knowledge/skills, 60% agreed that examination practice materials were easy to locate, 55% agreed that they were provided with good support in exam preparation, and 68% agreed that they had easy access to subject groups to help prepare for the exam.

Some of the resources sighted for the MCQ examination were as follows:

- MCQ exam information that provides detailed information on the computer-based exam process. Trainees can view this prior to sitting the exam (e.g. to ensure that they don't accidentally end the exam too early). It also includes examples of warnings they may receive.
- There is a practice exam in both computer-based and pdf format (the pdf has the computer instructions), which both familiarises them with the computer format and the question types, both MCQ (25) and CAP (1) for practice.
- Answers are provided for the EMQ and CAP, although no extra feedback is attached.
- A wide range of sample questions and answers for EMQs and CAPs.
- Past exam reports from 2014, which give brief summaries of candidate strengths and weaknesses on the written papers.
- A guideline for the type of knowledge that may be assessed in the CAP (2012). There is significant detail on the level of knowledge of these questions.

For the MCQ exam, the majority of respondents agreed that their trainer/supervisor had a good sense of their skill prior to the MCQ exam (66%); however, fewer agreed that their trainer/supervisor demonstrated good knowledge of the exam (45%). The majority of respondents agreed that their other program commitments suffered while they prepared for the exam (63%), whereas fewer agreed that their exam preparation suffered due to their program commitments (53%). The majority of respondents (75%) indicated that they spent 49 hours or more on practice MCQ questions prior to first attempting the MCQ exam; however, few agreed that the MCQ exam preparation support from the College met their needs (23% agreed). Only 25% of respondents used the resources on Learn*it* to prepare for the MCQ exam, with 40% of those who used these resources agreeing that they were helpful. Most

respondents used a private examination preparation course to prepare for the MCQ exam (83%) and most agreed that the course was worth the cost (89%). Only 22% agreed that there were good resources available to help develop the appropriate knowledge/skills, 36% agreed that examination practice materials were easy to locate, 55% agreed that they were provided with good support in exam preparation, and 66% agreed that they had easy access to subject groups to help prepare for the exam.

Some of the resources sighted for the Essay-style examination were as follows:

- A range of previous CEQs are available to assist with exam preparation. These include questions and scoring keys published for candidates (2016–2019 questions and scoring keys are available).
- A range of papers and scoring keys for the MEQs (including generic and mock versions from 2015–2019) are available for candidates.
- An e-module 'Guide to the CEQ' is available to trainees. This module presents the concept of the CEQ, explains scoring, shows samples of the CEQ to comment on, compares with examiner's comments, provides tools and methods to formulate the approach to the CEQ, reviews previous CEQ, and offers other resources to assist in CEQ response (duration 60 minutes).
- A guide to candidates on CEQ essay writing (a published journal article in *Australasian Psychiatry* 2017, 1–4).
- A reflection of a CEQ marker on what makes a good CEQ (*Australasian Psychiatry* 2018, 1–3).
- Past exam reports on essay style questions from 2016–2018 describing how candidates performed on individual questions in the exam that year, including candidate strengths and weaknesses.

For the Essay-style exam, the majority of respondents agreed that their trainer/supervisor had a good sense of their skill prior to the Essay-style exam (67%); however, fewer agreed that their trainer/supervisor demonstrated good knowledge of the exam (44%). The majority of respondents agreed that their other program commitments suffered while they prepared for the exam (70%), whereas fewer agreed that their exam preparation suffered due to their program commitments (61%). The majority of respondents (69%) indicated that they spent 49 hours or more on practice Essay-style exam questions prior to first attempting the Essay-style exam; however, very few agreed that the Essay-style exam preparation support from the College met their needs (17% agreed). The majority of respondents used the resources on *Learnit* to prepare for the Essay-style exam (63%), with 42% of those who used these resources agreeing that they were helpful. Many respondents used a private examination preparation course to prepare for the Essay-style exam (75%) and the majority agreed that the course was worth the cost (71% agreed).

Preconference workshops at the Annual Scientific Meeting run by the examinations committee were generally seen to also be helpful. It was also noted that candidates form study groups and source other material to use for examination practice. An annual mock examination has been available from New Zealand for several years. Some Australian states run their own mock essay examinations. In essence, there are lots of resources 'floating around'. It was generally agreed that the College did not need to step in to run more candidate support sessions as enough was being done. However, more collaboration between the training networks around making material available to other trainees and SMIGs, especially those placed in regional and remote centres, would be beneficial to trainees.

CANDIDATE REPORTING

Sample candidate results letters were sighted but only for the MCQ examination. Although candidates Pass or Fail the examination, the reporting used the less confronting language of being 'successful' or 'unsuccessful'. Candidates receive their percentage scores by examination component as well as by

blueprint domain. This is good practice and it allows the candidates to see their areas of strength and weakness. However, as the number of items in each category is not disclosed, the candidates cannot see whether the gap in their knowledge is substantial or has arisen due to a statistical anomaly. Also, if these domain scores do not account for differences in the average difficulty of items within each domain, then there is some risk that the feedback is misguided – for example, if a candidate appears weak on one domain, but that domain was the hardest, in reality they may not have a deficiency. There were also some concerns about the validity of the domain-level reporting due to the quality of the blueprinting.

The precise cut scores are not reported but candidates are given a visual indication of how close their scores were relative to the cut score. It was also noted that only unsuccessful candidates are given the visual indication. Successful candidates would also benefit from knowledge where they sit in relation to the cut scores. Although the current candidate reporting is adequate, there is room for some enhancement. Trainees seem to always want more when it comes to feedback, and there are often alternative ways for presenting as much information as possible in a way that is helpful and meaningful.

It was also pleasing to see that the College already provides reports to Directors of Training (with specific mention if a trainee requires targeted learning or to show cause), Branch Training Committees, reports at the State level, as well as a detailed post examination report from the Committee for Examinations.

FINDINGS: OBJECTIVE STRUCTURED CLINICAL EXAMINATIONS

PURPOSE AND FORMAT

The *Objective Structured Clinical Examination* (OSCE) comprises a short station circuit (comprising 8 short stations and 2 inactive byes, each of 10 minutes duration) and a long station circuit (comprising 3 long stations and 2 byes, which may be active or inactive, each of 20 minutes duration). The exams are offered twice a year. RANZCP recommends that trainees attempt the OSCEs during Stage 3 of training, but candidates may apply to complete this assessment after they have completed 30 months of FTE training, with at least 18 months in Stage 2.

The OSCEs are the only summative assessment of clinical skills. The purpose is to sample broadly and reliably across different curriculum domains to ensure that trainees are ready to practice as a general psychiatrist. In a high-stakes setting, the OSCE aims to provide an external assessment of clinical skill acumen. It aims to provide a robust way of ensuring that trainees are ready for independent practice by providing an opportunity to assess across a range of domains and interactions, and garner multiple examiner perspectives on a candidate. There were concerns raised that the large number of stations and short time available resulted in assessment of a very narrow spectrum of practice; indeed, each station only allows for a cursory assessment of psychiatric practice rather than a deeper probing of the art of psychiatry. These concerns are discussed in the next section.

STATION QUALITY & RUBRIC DESIGN

This section of the report summarises the overall quality of the OSCEs, based on the desktop review of submitted examination material. Eleven sample OSCEs were provided to ACER for comment. The RANZCP OSCE format is made up of two types of station: a role-play format in which the examiner(s) observe the candidate's performance for a given scenario, and a Viva-style format where the candidate directs their responses directly to the examiner. Since the OSCE format is based on successive and discrete stations, there is no inherent problem with using the format to deliver different station types, as long as the different expectations and required performances are clear to candidates. From a review of the supporting documentation, this certainly appears to be the case for the RANZCP OSCEs.

The primary value of the OSCE format is that it enables assessment of knowledge-based clinical skills in a safe, standardised and replicable environment, offering the possibility for good reliability for the assessment of such clinical skills. In terms of the popular assessment framework known as Millers Pyramid (Miller, 1990), the OSCE targets candidates' ability to *show how* they would respond in a given scenario. This type of assessment assesses candidates' 'maximum performance' in a controlled and structured situation (sometimes referred to as 'in vitro' assessment); while this performance will usually incorporate the lower levels of Millers pyramid (Knowing and Knowing How), it does not lend itself to strong conclusions about a candidate's *typical* clinical performance in real-life clinical environments (the so-called 'Does' level), due to the structured and highly controlled assessment environment of the OSCE. This higher level ('in vivo' assessment) requires an appropriate workplace-based assessment format. These considerations are intended to draw attention to the fact that while the OSCE tends to deliver high reliability, the validity of the assessment will depend on other aspects of assessment design and implementation, such as: the authenticity of the clinical scenarios, the degree of alignment with expected competencies (blueprinting), the clarity of instructions (to candidates, examiners and role players alike), training and calibrating of examiners, procedures for empirical standard setting, and the design of appropriate marking rubrics. While the structured and somewhat artificial assessment environment of the OSCE limits inferences that can be made about the integrated and overall clinical

competence of candidates in actual clinical environments, well-designed OSCEs provide invaluable information about key discrete competencies of candidates.

The above scope of assessment of the OSCE format is appropriately recognised in the RANZCP 'OSCE Policy and Procedure' document, which articulates the overarching aims of the OSCE as 'a predetermined range of practical skills', covering different areas and contexts of clinical psychiatry practice and aligning strongly with both Fellowship competencies and CanMEDS roles. The skills specified for assessment also appear appropriate and amenable to the OSCE format and are consistent with the kinds of skills typically assessed by the OSCE in medical and health education contexts (see, for example, Chapter 5 'What is assessed in an OSCE?' in Harden et al., 2016).

Finally, review of each station's clinical context from a lay educator perspective suggests highly authentic and well-thought out scenarios. More thorough review of the station contexts requires appropriate expert knowledge, and this appears to be built into the development process through strong station writer training and support, and a structured and documented review process.

The OSCE is traditionally marked, or scored, in one of two ways: through discrete 'checklist' items that identify performance/non-performance of pre-determined knowledge or skill elements; or by using rating scales that capture the quality and/or comprehensiveness of performance in particular domains of knowledge or skill. Checklists are usually reserved for skills that can be broken down into discrete elements whose execution is mandatory to the skill itself. This approach to scoring has the benefit of helping focus the examiners on the key elements to assess, to collect clear examples for feedback to candidates, and to score the performance systematically and objectively (Harden et al., 2016), contributing to the OSCE's generally high reliability coefficients.

Alternatively, rating scale scoring enables more holistic and integrative judgements of performance, while also allowing broader and multifaceted skills such as communication, collaboration, empathy, patient-centredness and organisational skills to be more appropriately assessed. Rating scales also more explicitly outline (for examiners and candidates) the different levels of performance pertaining to different stages of competence development. Given the nature of psychiatric clinical practice, and the skills listed in the Policy and Procedures document as assessed through the OSCE format, it would appear highly appropriate that the RANZCP OSCEs all use the rating scale approach to assess candidate performance.

However, potential disadvantages of scoring by rating scales include: proneness to examiner subjectivity and idiosyncrasy of judgement, lack of suitable clarification of criteria, and less transparency in terms of what constitutes below-standard performance (Harden et al., 2016). These disadvantages can be mitigated to some extent through explicit criteria, examiner training and psychometric review of results, but there is little doubt that rating scales tend to allow more examiner variability than checklists.

The two scoring approaches can be used side-by-side, if a particular station warranted parts of the performance to be assessed by way of checklist, although this may not be compatible with the current alignment with CanMEDs competencies and proportional weighting of these elements. On the basis of the supplied samples, we see no great need to introduce checklist scoring at this point, unless reliability data suggested a significant decline in the reproducibility of examiner scoring on the rating scales (in which case, this would indicate that the task or rubric likely needs work). These considerations are ultimately a trade-off between reliability and validity; in our view, validity is key, but requires *sufficient* reliability to support valid inferences about competence. In other words, reliability for its own sake is counter-productive, and usually comes at the cost of some validity. Therefore, sufficient reliability is defensible if the assessment is designed appropriately and forms part of a well-planned and structured programmatic approach to assessment.

The RANZCP OSCE scale

All RANZCP OSCEs reviewed used a 6-point scale with four levels of performance: Surpasses Standard (scored as 5), Achieves Standard (scored as 3 or 4), Below the Standard (scored as 1 or 2), and Domain Not Addressed (scored 0). Each level is explicated with a description (varying in specificity) of the relevant behaviour or expected performance of candidates, in line with the provided scenario and the relevant domain.

The highest score category (surpasses) is generally well-described using a range of descriptors that signify exceptional performance, e.g. comprehensive, sophisticated, superior. While some subjectivity remains in these terms, they are generally well-understood and appropriately rely on the examiners' experience and judgement for this highest level of performance, which clearly exceeds the minimum standard. The minimum standard (score category 3) appropriately has the most explicit and detailed descriptions and reflects most directly the specific features of the given clinical context. This is good practice given the high-stakes nature of the OSCE and the fundamental importance of the *At Standard / Below Standard* distinction. The clear attempt to tailor the descriptions of minimum standard to the specific scenario is highly commendable and would appear to significantly support the validity of such decisions. In fact, the second element of the Achieves Standard descriptors (i.e. beginning with the statement 'To achieve the standard ...') aligns well with level 3 of the previously mentioned marking guide taxonomy (Ahmed & Pollitt, 2011), in that it provides a clear rule or principle, specific to the clinical scenario, to differentiate between at standard and below standard responses. This principle approach recognises that candidates may achieve the standard in slightly different ways, while still meeting this core criterion.

However, this approach is somewhat inconsistent with the more detailed description of the expected performance at this level. For example, Station 4 (Medical Expert domain) reads as follows:

Achieves the Standard by:

demonstrating use of a tailored biopsychosocial approach; assessment process is hypothesis-driven; obtaining a history relevant to the patient's problems and circumstances with appropriate depth and breadth; seeking previous psychiatric / medical history and presentation; repeated examinations of orientation because of possible fluctuation in presentation; identifying relevant investigations; demonstrating ability to prioritise assessment of risk and safety (e.g. absconding risk; 'sun downing'); screening for delirium and dementia syndrome; and outlining the key issues including cognitive examination.

To achieve the standard (**scores 3**) the candidate **MUST**:

- a. Explore the chronic nature of the visual perceptual abnormalities.

As mentioned above, the second part is a good example of a principle-based marking approach. However, the preceding detail appears to present alternative criteria for achieving the standard; some appear consistent with the stated principle, while others seem quite different. It is not clear to us which of the two criteria for 'achieving the standard' the examiner is expected to follow. It seems entirely possible for a candidate to meet one or more of the detailed criteria but not the stated principle. Perhaps this is more fully explained to examiners elsewhere. The greatest risk here is that, in our experience, in the face of contradictory criteria, examiners frequently decide on their own version or reading of the criteria as the basis for their judgements, with implication for the validity of the task. This discrepancy seems to apply to most descriptions of the Achieves Standard level across the stations.

Other descriptors for score categories 1, 2 and 4 appear to be common across the stations and are written in more general terms. This also seems appropriate, due to the lower stakes of these score points (example below).

A score of 4 may be awarded depending on the depth and breadth of additional factors covered; if the candidate includes most or all correct elements.

Below the Standard (scores 2): scores 2 if the candidate does not meet (a) above or has omissions that would detract from the overall quality response.

Below the Standard (scores 1): scores 1 if there are significant omissions affecting quality; omissions adversely impact on the proposed content to be obtained; significant deficiencies such as substantial omissions in history or assessment process

However, here too there are some inconsistencies. Both score categories 1 and 2 refer to omissions that impact on 'quality'. It is not clear how the accompanying text is intended to help differentiate between these two scores. This appears to leave a great deal to individual examiner judgement, and could conceivably be applied differently between examiners and within examiners. Similarly, there is a potential inconsistency in category 4 between 'depth and breadth of additional factors' and 'most or all correct elements'. Which is the examiner's judgement to be based on, qualitative differences in performance (breadth and depth) or a count of correct elements? While the score points around the minimum standard are clearly not as crucial for determining the pass/fail decision on a particular OSCE, they may impact overall decisions when summated. Therefore confusion or inconsistency around these categories can still significantly impact on candidate results.

Overall, however, the OSCEs appear to be appropriately used and written, providing authentic clinical contexts and tasks for candidates, with relatively clear marking guides tailored to both clinical context and domain. There appears to be a significant inconsistency within the Category 3 'Achieves Standard' descriptors that should be clarified, and a couple of relatively minor discrepancies in other score categories that should be at least reviewed. The exclusive use of rating scales for scoring performances appears justified by the nature of psychiatric clinical practice and the particular skills being assessed, as long as the reliability of the results remains adequate. However, there are implications for the current approaches to setting cut scores, which are discussed shortly.

CONSTRUCTIVE ALIGNMENT

The OSCE blueprint is constructed with reference to CanMEDs competencies and by topic, with reference to previous examinations in order to limit duplication. Although we noted that station criteria are blueprinted to CanMEDs, we did not sight any more detailed mapping, such as topic areas within Medical Expert by station criteria. Thus, blueprinting for the OSCEs is not as detailed as it should be.

As mentioned in the Written Examination section above, blueprinting is a process that is part of a broader program assessment framework. Although it appears that blueprinting processes exist for the OSCE, no assessment framework was sighted. Although stations are mapped to Blueprint Descriptor Categories, Areas of Practice, CanMEDs Domains and Fellowship Learning Outcomes, there was no overarching framework that provided guidelines to the subcommittee around appropriate coverage of content and other domains in any one administration.

FITNESS-FOR-PURPOSE

Fifteen members of the Committee for Exams responded to questions regarding the quality of the OSCEs in the survey. On a scale from 0 to 100, the mean ratings were as follows: quality of the OSCE content (71.8), validity of the OSCE format (63.0) and usefulness of the OSCE score (59.1).

Trainees and SIMGs generally responded positively to questions related to the OSCEs. Respondents agreed that they had adequate time to prepare for the OSCEs (85%), that the curriculum clearly identifies what needs to be learnt for examination success (80%), that they had a good understanding of what is assessed in the exam (83%), and that the skills assessed in the exam were well aligned with

their training program (79%). Most respondents agreed that the OSCE was difficult (84%), but most felt that they spent enough time preparing (95%), were well prepared for the format of the Essay-style exam (94%), and that their skill level was right for the exam (97%).

The majority of Directors of training and supervisors believed that they had a good understanding of what is assessed in the OSCE (96% agreed). In general, most respondents agreed that the skills assessed in the OSCEs aligned well with the training program (95%), that the format of the OSCE was important for the trait being measured (90%), and that it was important to assess the traits measured in the OSCE (96%). Most respondents agreed that they had a good sense of trainee skill-level prior to them attempting the OSCE (94%) and they were confident that most of their trainees who attempted the OSCE were progressed enough to pass it (85%). Fewer respondents agreed that the OSCE results provided to candidates were useful (72%), that the OSCE results helped them identify trainees' strengths and weaknesses (66%), that they could identify a trainee's skill level from their OSCE result (58%), or that they used OSCE results to evaluate their training programs (55%) or identify training resources needed (64%).

Historical perspectives and the OCI

Although the RANZCP OSCEs appear fit-for-purpose, there was widespread disquiet that the OSCEs were not assessing the core skills of psychiatrists. This speaks to a deeper philosophical question about the fitness-of-purpose of the OSCEs and whether the purported purpose is valid. It is our view that as they stand, the OSCEs are fit-for-purpose in that they are clearly assessing what OSCEs are supposed to assess. However, this does not mean that all core areas of psychiatric practice are currently assessed by the OSCE.

One theme that arose overwhelmingly throughout the consultations was that core skills around meeting and communicating with real patients, taking a patient's history, conducting a patient examination, making a diagnosis and formulating a treatment plan, are not being examined. In the past, these skills were assessed summatively in an Observed Clinical Interview (OCI). Sometime around 2014, the OCI was formally removed from the examinations, and these skills are now only assessed in an Observed Clinical Activity (OCA) as part of the workplace-based assessment program. There was strong consensus that this move has negatively impacted the training program. Assessment drives learning, and trainees spend their greatest efforts learning for the high-stakes assessments (i.e. examinations). It was agreed that although trainees may have dreaded the OCI, it was a strong driver of learning the core skills of a psychiatrist. As there are no longer any stakes attached to the OCA, supervisors are more likely to 'sign candidates off' so that they can progress through training. The assessment is designed to be formative, but the general consensus was that this task is not working well. Although the workplace-based assessments are outside the scope of this review, the relevance of this issue has caused us to make some general remarks about their place in the entire assessment program in the conclusion of this report.

We understand some of the psychometric challenges of an OCI, in that the assessment would have low reliability. Indeed, in order to meet statistical thresholds for reliability, the number of stations may have been unfeasible. However, when considering the utility of an assessment instrument, reliability is only one factor. Assessment is an optimisation problem, and there are always trade-offs. Van der Vleuten (1996) defined a framework for evaluating the utility (U) of a given assessment: $U = \text{reliability (R)} \times \text{validity (V)} \times \text{educational impact (E)} \times \text{acceptability (A)} \times \text{cost (C)}$. It seems that the OCI was the most authentic assessment of psychiatric practice, and it forced trainees to practice the core skills of their profession. So, although the reliability of the OCI was low (and the costs were potentially high), the validity and educational impact would have been extremely high, underscoring the value that would have been added by the OCI.

Researchers and practitioners in the assessment in medical education landscape are no longer striving for objectivity in assessment above all else. There is a shift towards embracing subjective and qualitative data in appropriate contexts (see Hodges, 2013; ten Cate & Regehr, 2019). There is a notable tension, in that workplace-based assessments are generally seen as less reliable. But in specialty training contexts, it is generally acknowledged that a black-and-white distinction between formative and summative assessments results in trainees focusing overwhelmingly on the summative assessments and de-valuing the formative assessments. The solution to this predicament is to build a continuum of stakes into the assessments and make high-stakes decisions on trainee progression by accumulating data from lower-stakes assessment moments. A programmatic approach to assessment (see Schuwirth & van der Vleuten, 2011) may be helpful here, and there are specific guidelines for incorporating elements of programmatic thinking into a speciality training context that may provide a way forward, such as accumulating assessment information from the OCAs to inform trainee decision-making about trainee progression (see Pearce & Prideaux, 2019).

FINDINGS: OBJECTIVE STRUCTURED CLINICAL EXAMINATION PROCESSES

STATION DEVELOPMENT

The OSCE Committee aims to select a range of topics and competencies across the blueprint, although the Medical Expert competency is a primary focus of the OSCEs. OSCE topics are selected and assigned to primary writers who work with a colleague who is responsible for editing the OSCE.

Development of the OSCE begins 12 months in advance of the examination sitting and with reference to the OSCE blueprint. Topics are assigned to a large group of station writers who are also given a 'buddy' to work with on development. Development begins individually and is guided by an OSCE template. After approximately 6 weeks, the draft is sent to the buddy to edit, as well as the chair and deputy chair of the committee. The development process is collaborative and iterative and goes through multiple versions prior to completion. Twice a year, stations are tabled and thoroughly reviewed by other members of the subcommittee. Very occasionally an expert outside the OSCE committee may be consulted for feedback on a particular station. The final version is typically ready at least two months before the OSCE. Some minor final editing may occur after this closer to the examination itself.

The marking rubric design has evolved organically over time. Specific detail is given to aligning the stations with learning objectives and the CanMEDs framework, although the stations are still heavily focused on the Medical Expert domain. The detail in the rubric is developed along with the station. More recently, writers have been asked to include 'key responses' that identify the core elements of proficiency a candidate 'must' do for the station to be minimally competent. There is some degree of flexibility here and, in line with good practice, the rubrics are designed to be task-specific rather than generic. On occasion, the number of elements that the candidate must demonstrate was regarded as excessive, with concerns that some elements were pedantic. Overall, however, the collaborative and iterative processes for OSCE station development appear robust.

EXAM DELIVERY

Delivery of the OSCEs occurs twice a year in two different locations. These examinations are an extremely complex exercise involving significant numbers of examiners, candidates and College staff. The delivery of the OSCEs in Perth in September 2019 utilised two venues (Royal Perth Hospital and Fiona Stanley Hospital) with data processing occurring on site at each venue. The operation of this OSCE at the Royal Perth Hospital site ran very smoothly.

The OSCE circuits are clearly labelled for candidates and examiners and there are printed maps for examiners to assist orientation. Timing and candidate flow between stations is managed efficiently by College staff and errors (such as candidates inadvertently skipping a station) are extremely rare and quickly rectified. Examiners, candidates, College staff and observers are all clearly identifiable through different coloured lanyards. Observers observe OSCEs throughout the day and although there is no formal observer report, there may be informal discussion about marking between observers and examiners. In some respects, the hospital environment is well-suited to the OSCE, with many available rooms in the outpatient department. However, the noise from candidates and role players in adjacent rooms is clearly audible on occasion. For this reason, examiners are instructed to remain quiet between candidates and refrain from discussing previous candidates, which all adhered to during the day.

The AMC testing centre has been suggested as a possible alternate venue for RANZCP OSCEs. This centre would certainly have the advantage of being purpose-built to deliver clinical examinations and would enable video-recording of assessments for the purpose of training and candidate review. However, it would also mean that large numbers of candidates from around Australia and New Zealand would be compelled to travel exclusively to Melbourne to complete the OSCEs. The variation in location of the OSCEs would seem to be an important part of ensuring equity of access for candidates and in involving examiners from across Australia and New Zealand.

Regardless, the benefits of video-recording clinical examinations may outweigh the negatives. Alternative arrangements for filming examinations could be explored. As well as enhancing options for training and calibration, recording examinations ensures that objective evidence exists in the case of a legal challenge. This material can be used to defend the processes of the College or the professional judgment of an examiner. It was noted that currently, appeals rarely progress as there is insufficient evidence. However, increasingly, litigation is happening throughout the medical specialist training landscape and the College needs to at least be aware of the implications of having no video evidence.

The number of trainees applying to undertake the OSCEs is increasing. Increasing the number of sittings of the OSCE to three is unfeasible because of the increase in resources and the burden on volunteers to develop OSCEs for three examinations per year. Other challenges that arise because of the increase in candidate numbers include the logistical challenge in allocating examiners to candidates who are sitting multiple times. The College attempts to minimise perceptions of bias by ensuring that examiners do not re-examine a candidate who they have previously failed. The College may need to implement a cap on the number of candidates in the future. This may be best achieved by placing a limit on the number of opportunities trainees have for the OSCEs, as well as reforming the progression pathways so that trainees only attempt the OSCEs when they are well and truly ready. This topic is further considered in the conclusion to this report.

MARKING PROCESSES

For the three 20-minute long stations, two examiners independently mark according to the marking rubric. Each of the remaining OSCEs is marked by a single examiner. At the end of each candidate, the examiners enclose their marking sheet in an envelope and these are collected by College staff. Candidate data is scanned and entered throughout the OSCE, which makes it possible to follow up with outlier examiners who employ highly inconsistent marking patterns (e.g. overuse of the zero category). However, the complete data file requires some weeks of checking following the OSCE and is not available for discussion by examiners at the end of the day.

While double marking for all stations would be preferable to minimise examiner bias, it would be pragmatically very difficult to achieve with the current number of stations and candidate numbers. If numbers reduce in the future due to other reforms, double marking options should be explored. It would

also be worth exploring the options for moving to an electronic (such as a tablet) marking platform. This would streamline the marking process and reduce the potential for errors.

PSYCHOMETRIC ANALYSES

A review of all submitted examination data was conducted by ACER psychometricians. For the OSCE, we were additionally able to draw upon our involvement with data analysis and checking during April and May in 2019, including our participation in the standard setting committee meetings for this examination.

Scoring of the OSCEs

Examiners' raw ratings (on the 6-point OSCE scale) are collated for the 11 stations along with their global proficiency rating for each station. Each station has up to 5 criteria, which have a proportionality weighting. The criteria are blueprinted to CanMEDs domains: Medical Expert, Communicator, Collaborator, Manager, Health Advocate, Scholar and Professional. The raw ratings are translated into criteria scores, based on the proportionality weightings, and a final score out of 50 is calculated for each station. The candidate's total scores are the sum of their station scores. The cut score for each station is calculated as the mean of the scores that have a 'marginal performance' global proficiency rating, and the sum of these is the overall cut score for the examination. This is discussed in the review of standard setting below.

Nevertheless, it is worth flagging that the rationale and processes for converting domain ratings (from a categorical scale) to numerical scores is somewhat questionable. From a measurement perspective, multiplying the scores by proportionality factors to arrive at a final score out of 50 seems arbitrary, and runs the risk of losing the essential assessment information contained in the examiner ratings associated with assessment domains. Aggregating data across domains to interrogate patterns of performance would be a worthwhile exercise.

Station analysis and review

Detailed performance of the stations is prepared by College staff for review by the committee. This includes analyses of station mean scores and standard deviation by stream, and comparisons of performance between trainees and SIMGs. Overall examiner statistics are also prepared for discussion at a post-OSCE teleconference, interrogating inter-examiner consistency. These are presented along with statistics on overall pass/fail numbers and rates broken down by different categories. It would be worth including distributions of candidate scores by station as well. Overall, these analyses and the opportunities they provide for discussion are good practice. It was noted that the statistical analyses are fed back to the committee to inform development of new OSCE stations. However, the statistics generated based on station scores of 50 are multiple steps removed from the raw ratings of examiners. This means that careful data interpretation is required by the committee to safeguard against spurious conclusions being reached based on the data.

STANDARD SETTING AND CUT-SCORE DETERMINATION

As noted in the findings for the written examinations, standard setting is a process of systematically determining the score on an examination which represents the minimum acceptable level of performance. Standard setting is a more defensible approach than determining a pass mark (or level of performance) by default, custom or arbitrarily. Essentially, the standard setting process attempts to define what level of performance on an exam is to be regarded as 'just good enough' for the current stage of a learning programme or for certification of a defined level of competence.

Performance-based methods rely on actual performance in the exam, whereby data on the performance of minimally competent candidates is subsequently used to calculate an appropriate cut score.

Performance-based approaches are often utilised for clinical exams (including OSCEs); however, they require a minimum number of candidates per exam in order to produce reliable cut scores, normally in excess of 50 performances. Fortunately, the candidate numbers ensure that performance-based methods can be employed for the RANZCP OSCEs.

The standard setting approach employed for the OSCEs is a version of the Contrasting Groups or Borderline Groups method (see Zieky & Livingston, 1997). The actual test scores of a borderline group are obtained, and the mean of this distribution becomes the recommended cut score. This is how the Global Proficiency Rating data is utilised. The three possible groups used by RANZCP are 'Definite Pass', 'Marginal Performance', and 'Definite Fail'. The mean scores of the 'Marginal Performance' group of candidates becomes the cut score for each station, and the overall cut score for the OSCEs is the sum of these scores.

The current model for determining the overall cut score is a compensatory model. This focuses on overall performance, aggregating scores across all stations in the exam. This allows for candidates to fail in some stations, and indeed compensate these failures with passes elsewhere. Compensatory approaches are often contrasted with conjunctive approaches, whereby minimum standards are set for each specific station or a specified proportion of domains or components in an exam. For example, a conjunctive approach would involve setting a rule (and clearly communicating the rule to candidates) that candidates must meet the acceptable standard in 8 out of 11 stations. The specifics of a conjunctive approach are ultimately a policy decision, and it depends on how the curriculum is conceptualised and structured.

The current approach is a valid and defensible performance-based method for setting the cut score on OSCEs. However, its precise implementation for the RANZCP OSCEs requires some careful reconsideration. Typically, this method is employed to convert a conceptual minimal performance standard to a cut score on a test. But this assumes that the test score is a valid measure of performance. As already highlighted, because we are concerned that as the raw categorical data from the marking rubric design has already gone through a convoluted and zealous translation process to arrive at the final score, the validity of the standard setting process may be compromised. In fact, this points to a strength of the current OSCE marking rubric design, in that the stations have detailed descriptors for examiners around minimal levels of acceptable performance. The rubrics are tailored with descriptions of minimum standards related to the specific station scenario (substantial training and calibration occurs, as discussed later), and minor differences in station difficulty are already accounted for in some sense. The principle approach in the rubric design coupled with the essential elements of the task (that the candidate 'must' achieve) points to robust consideration of performance standards in developing the rubric and thus in examiners deploying their judgment to rate candidate performance. Examiners are not given analytic scores without consideration of the standard that needs to be later determined. Instead, examiners are rightly making judgments about candidate performance with respect to a conceptual performance standard with task-specific guidance each time they rate performance in an OSCE. We are concerned that this crucial detail is being lost through the current scoring and standard setting process.

Perspectives from stakeholders

Some of our concerns were echoed throughout the consultations. Overall, examiners were concerned about the way the standard setting is implemented for the OSCEs. Although some expressed concerns about the borderline group method itself, it appears that most of the disquiet arose due to perceptions that candidates were passing overall who should not be passing. This is the problem of false-positives – if standards are set too low through the standard setting process, candidates will pass even though they are not yet safe to practice. Examiners are often stunned that, after the College analysis, so many

candidates have passed when so many of them were below standard in so many stations. Many were concerned that this was a function of how the standard setting method worked.

This method hinges on examiners' understanding and application of the global proficiency rating. It was clear that many examiners struggle to use the global proficiency rating and some wanted it scrapped altogether. Examiners may be inclined to 'sit on the fence' and rate a candidate's performance as marginal, perhaps also due to the strict wordings of '*definite pass*' and '*definite fail*'. If examiners select 'marginal performance' for many candidates who are below standard, this could artificially lower the cut score. Also, review of many of the cut scores suggests that these were similar to the average performance of the station, which suggests that examiners are not differentiating between a marginal and an average performance.

Options for consideration

In reviewing the rubric design, we had expected to see a conjunctive approach to standard setting. The holistic approach with its task specificity suggests a model whereby minimum standards need to be met across the curriculum and blueprinted domains. As well as aggregating examiner ratings by station, the data could be aggregated in other ways. Doing so would allow for detection of candidate strengths and weaknesses in specific areas, such as patient communication, data gathering, or making a diagnosis. Patterns of proficiency may emerge across stations. Currently, candidates may be meeting the overall cut score but performing consistently below the acceptable standard in certain blueprint categories. For example, it is conceivable that a candidate may get every diagnosis wrong in the OSCE stations but still pass overall by compensating with their other abilities. Aggregating data by assessment domain as well as by station would allow for conjunctive pass/fail rules to be established by content or skill, and would be an example of incorporating programmatic thinking into examination data (see Schuwirth & van der Vleuten, 2011; Pearce & Prideaux, 2019).

We also note that the three longer stations are equally weighted with the 8 shorter stations. It may be worth aggregating data differently between the longer and shorter stations. As well as aggregating data across stations to determine conjunctive rules based on blueprint domain, conjunctive rules could be established for having to pass a certain number of stations. Both of these approaches could be trialled using existing examination data to ensure that an empirical evidence-base informs the development, and the impact on the cohort should be analysed prior to finalising and publishing conjunctive rules. In doing so, it is important to remember that standard setting contains a substantive procedural aspect, rather than blind application of a methodology. The choice of method requires credibility, and must be feasible and defensible in the literature. Due diligence must be applied to the way it is implemented. The approach taken must have a sound rationale, be clearly articulated, be consistent and be systematic. Finally, it is important to consider whether the results of the standard setting procedure are fundamentally fair.

BORDERLINE CANDIDATE REVIEW

After an examination sitting, the committee reviews detailed results in a teleconference with support from the College. Detailed performance of the stations (discussed earlier) is prepared by College staff for review by the OSCE subcommittee. The committee review the specific results of every borderline candidate. Rarely, a 'conceded pass' may be granted if a candidate is very close to the cut score and there is sufficient evidence that they were proficient overall. However, typically the borderline candidates have been deemed to be below the minimum standard in many of the 11 stations.

As already mentioned, due to the implementation of the standard setting and the compensatory nature of the process, candidates have often failed several stations but are still close to passing overall. Although we did not attend one of these review meetings as part of this review, through reviewing the submitted documentation and talking through the processes with stakeholders, the process for

borderline candidate review appears robust and thorough. As with the written examinations, upon identifying borderline candidates, deliberations by the panel should focus on available data using an evidentiary framework. That is, where small deficiencies in performance in certain domains are identified, other evidence from across the assessment program could be used to inform decision-making.

TRAINING AND CALIBRATION OF EXAMINERS

While there is no formal training pathway for new examiners, informal training and induction does occur. For example, new examiners are paired with experienced examiners for the long stations when they first participate in the examinations. Furthermore, an impressive amount of work goes into the training and calibrating of examiners prior to the OSCE.

Two months prior to the OSCEs, videos of the stations are recorded for training purposes. The videos are generally pitched at a 'borderline performance', which can be challenging for examiners to execute. The author of the station typically examines in the video, the buddy is the role player, and another committee member plays the candidate. Feedback is given by other members and the Chair and Deputy Chair who watch every station being taped. This exercise adds a further quality check on the station itself.

On the day before the OSCEs, these videos are used to orient role players to the relevant station and their role. Examiners for the same station gather together following the role player training and are briefed on expected examiner behaviour (e.g. scoring processes, observing silence between candidates, not collaborating on marking for long stations). Examiners have the opportunity to ask questions and to familiarise themselves with the content of the OSCE, including the 'must have' elements of the performance. Examiners are then further oriented to the content of the station and watch the relevant video performance. Following the video viewing, the OSCE author leads a calibration exercise with the group of examiners. The examiners are asked to suggest how they would score the global proficiency rating and examiner judgements are used as the basis of discussion on the performance standard. As a group, the examiners then suggest how they would each score the different competencies comprising the OSCE. The perspectives of examiners with contrasting views on the performance standard are used to generate discussion as to the appropriate rating. On occasion, it was not clear that the performance demonstrated in the video was aimed at the marginal level. In order to further familiarise examiners with the standard, a second performance that was clearly at the satisfactory level might also be useful. In the sessions observed, the calibration exercise was done as a group after the video was played. It may also be useful to ask examiners to practice-mark the OSCE using the rubric while they watched the video. The training exercise also includes further practice in small groups with the role player undertaking their role, and examiners playing the roles of candidate and examiner. This exercise provides an opportunity for the examiners to further understand the difficulty of the case, the likely responses of candidates and what is achievable in the time frame. These sample videos are also a valuable resource. It is possible that they could be scripted and re-filmed with actors, rather than examiners, to further contribute to examiner training.

Overall, the calibration work done the day before the OSCEs aligns with best practice. There has been some talk of facilitating this online, but we would caution against this. The face-to-face element ensures high engagement and allows for any issues with the stations to be identified prior to the examination the following day.

SUPPORT FOR CANDIDATES

Before undertaking their examinations, potential candidates are given access to a wide range of resources to help them prepare. Trainees have access to this material through their Learn*it* system. The following resources related to the OSCE were sighted:

- Candidates have access to a range of previous OSCE stations (substantial number of samples from 2011 to 2019 (April), including a full range of information for the stations that are available. As well as the recent OSCEs, there are some much older ones (back to 2006).
- Information regarding the way that the modified OSCE circuit works and a document outlining the rationale for this.
- A presentation on how to approach the OSCE.
- The description of the mapping of OSCEs to the OSCE blueprint descriptors. Eight disorder groups and 3 skill areas relevant to psychiatry.
- A document explaining the marking of OSCEs and the blueprint and the cut-score process (borderline groups).
- Detailed Examiners' Reports from the Committee for Examinations are made available to candidates. These reports include summaries of the stations, results summaries (including cut scores), means scores and pass rates, and some general feedback.

For the OSCE, the majority of respondents agreed that there were good resources available to help develop the appropriate knowledge/skills (87%), that examination practice materials were easy to locate (91%), that they were provided with good support in exam preparation (80%) and that they had easy access to subject groups to help prepare for the exam (78%). Most respondents agreed that their trainer/supervisor had a good sense of their skill prior to the OSCE (86%) and most also agreed that their trainer/supervisor demonstrated good knowledge of the exam (83%). The majority of respondents agreed that their other program commitments suffered while they prepared for the exam (64%), whereas slightly fewer agreed that their exam preparation suffered due to their program commitments (53%). The majority of respondents (75%) indicated that they spent 49 hours or more on OSCE practice prior to first attempting the OSCEs, with the majority agreeing that the OSCE preparation support from the College met their needs (68% agreed). Approximately half of all respondents used the resources on Learn*it* to prepare for the OSCE (53%), with 72% of those who used these resources agreeing that they were helpful. Many respondents used a private examination preparation course to prepare for the OSCEs (50%) and the majority agreed that the course was worth the cost (79%).

It was noted that support for trainees in preparing for the OSCEs would vary depending on their location. Rural and regional trainees have less access to face-to-face support, especially not being able to draw on the advice from a large number of supervisors. Nevertheless, it was generally acknowledged that the trainees have good study networks. More collaboration between the training networks around making material available to other trainees and SMIGs, especially those placed in regional and remote centres, would be beneficial to trainees.

Finally, the processes for being recommended to sit the OSCE do not appear to be working efficiently. That is, there was an indication that some supervisors allow their trainees to sit the OSCEs even if they believe the trainee is not yet ready. This ties in to the fact that there is no limit to the number of attempts available for trainees to sit the OSCE. We address this issue in the conclusion.

CANDIDATE REPORTING

Sample candidate results letters were sighted but only for the OSCEs. Although candidates Pass or Fail the examination, the reporting used the less confronting language, that 'you have/have not demonstrated the required standard'. A link to the OSCE Stations and Examination Report available on the College website is also included in the letters. Candidates receive their specific scores by station

as well as by blueprint domain. This is good practice and it allows the candidates to see their areas of strength and weakness. It was pleasing to see this level of detail provided to all candidates, even those who pass. However, as the data points in each category are not disclosed, the candidates cannot see whether the gap in their knowledge is substantial or has arisen due to a statistical anomaly. The blueprinting review noted that most of the assessment data relates to the Medical Expert domain. Therefore, a score of 0.0 on 'Health Advocate' with a mean cohort score of 23.1 may arise because of a small sample of Health Advocate data points.

Trainees seem to always want more when it comes to feedback, and there are often alternative ways for presenting as much information as possible in a way that is helpful and meaningful. Nevertheless, the current detail in the OSCE results letters do, from our perspective, reflect best practice. The way that this feedback is used to aid learning in training is a bigger issue. It may be unhelpful to a Director of Training to see that the trainee 'does not know enough about being a medical expert', for example. However, the detail in the letter coupled with the information in the examination report does provide reasonable and sufficient information for trainees.

Currently no further specific feedback on their performance is given to trainees. It would be difficult for examiners to fit this task into the limited time they have between candidates during the OSCE; indeed, this time may need to be increased. The provision of more specific feedback may also be feasible going forward if marking moved to a tablet platform. Examiners could spend 1 minute making some specific notes that may help the candidate in their future learning.

It was also pleasing to see that the College already provides reports to Directors of Training (with specific mention if a trainee requires targeted learning or to show cause), Branch Training Committees, reports at the State level, as well as a detailed post examination report from the Committee for Examinations.

OTHER FINDINGS RELATING TO THE ASSESSMENT PROGRAM

Although the non-examination components of the assessment program (including the curriculum and training itself) were outside the scope of this review, many issues relating to the training program and other assessments kept surfacing. Across all colleges that undertake medical specialty training, workplace-based assessments are often under-valued by trainees and trainers alike. There are concerns relating to variability in assessors, different experiences of the supervisor–trainee relationship, consultants who are reluctant to complete the necessary paperwork, difficulties finding time for the assessments, and forms filled in retrospectively ‘at the last minute’.

For many colleges, trainees who fail examinations have often done poorly in their formative assessments, but no intervention was undertaken to rectify their gaps in knowledge and ability. These assessments should play the role of identifying gaps in trainee learning early, so that plans for addressing deficiencies can be implemented early. It appeared that although these assessments were developed to serve this function, it appears that they are currently not appropriately fulfilling this role. This points to a clear disconnect between the relevant examination and training committees.

The clear-cut distinction between purely formative and summative assessment is unhelpful and perpetuates a false-dichotomy. Programmatic thinking that bridges the gap between training and assessment should be incorporated into the RANZCP assessment program. All assessment has the potential to generate information that can be aggregated and accumulated to form an evidence-base around trainee learning progression. Programmatic assessment treats assessment as a whole program, instead of isolated instruments (Schuwirth & van der Vleuten, 2011; Prideaux, 2016). The strength of an assessment program is thus judged by the program as a whole, rather than by the properties of single instruments. This links to the earlier discussion around the psychometric challenges of the OCI, and considering assessment as an optimisation problem.

Although the current RANZCP training program is articulated as a ‘competency-based’ fellowship program, the training pathway appears to be functionally time-based in that progression occurs automatically based on number of months in FTE training. This does not align with the fact that trainees learn and progress at different rates in different domains. The examinations appear to be driving trainee learning, as they automatically become eligible for these after fixed times. It would be worth exploring options for a hybrid model of training that implements competency-based hurdles, ensuring that trainees progress to the next phase of training when they have demonstrated adequate competency.

For example, many stakeholders believed that some trainees were sitting their OSCEs too early in training, before they had reached the required level of competency. This impacts the long-term sustainability of the OSCEs, particularly as candidate numbers are continually increasing. As the OSCEs are such a resource intensive exercise, it would be wise to develop a robust mechanism for ensuring that candidates do not sit until they have sufficient professional experience and have demonstrable attainment of the fundamentals of early training.

Enforcing a strict upper limit on the number of attempts or opportunities a trainee has at their final clinical exams (as enacted by multiple colleges) would also be worth considering. Although such reforms go beyond examinations and relate to broader training issues, because they would impact the feasibility of the examinations and allow for targeted specific reform to be implemented, we have included these suggestions in our recommendations.

CONCLUSION

Throughout this Review Project we have attempted to interrogate all aspects of the RANZCP Examinations in a thorough and robust manner. High-stakes examinations are sensitive to external review, and we were grateful for the high level of engagement and openness we experienced from RANZCP stakeholders. For the great majority of individuals involved at different stages of the examination process, there was a strong commitment to educational improvement across many areas.

The relevant committees and the College should be commended on their commitment to developing and delivering high-quality examinations. Many aspects of the examination processes are robust, with thorough quality controls built in. For this reason, we have given several commendations (see executive summary). Feedback from the consultative process was generally positive regarding the examinations and examination processes. Special appreciation should be made for the sheer volume of the work undertaken by volunteers on the different committees. The bulk of this work is 'unseen' and the College could look at ways for enhanced recognition of the time and effort that goes into the examinations.

However, in their current form, several existing processes require modifications to align with best practice. Examination processes have evolved organically over many years, generating somewhat convoluted implementations of certain methodologies. Overall, we argue that examination processes for RANZCP need to be streamlined and simplified. At present, there are many areas that would be subject to scrutiny and pose a risk to the integrity of examination processes at the College. These have been highlighted throughout the report and form the basis of our recommendations.

While there was general agreement that the majority of candidates who pass the Fellowship Examinations are safe to practice at a basic level, there was also a substantial issue identified: many felt that the three examinations together do not assess all core psychiatric skills. While each instrument may be technically sound, the focus of the examinations may not be completely aligned with the learning outcomes of the training program. These skills are possibly assessed outside of the current examinations, but assessing them in a lower-stakes context can impact trainee learning.

Our recommendations are detailed in the executive summary of this report. They identify modifications that can be made to the existing examinations in order to reflect best-practice in assessment. In the context of high-stakes examinations increasingly being subjected to scrutiny, we encourage the College to carefully reflect on these suggestions and take them into account.

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